THRIEND.IO: Meta Threads Social Media Platform Management and Automation

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Supervised By: Dr. Umer Rashid

Submitted By: Mohammad Shamil Khan

Department of Computer Science

Quaid-i-Azam University Islamabad

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ABSTRACT

In today's world social media platforms have significant importance and It's also necessary to maintain their profiles and accounts consistently with quality posts and engagement. And for this it requires a lot of time to manage profiles on daily basis to open Threads and post daily and engage with audience on daily basis. As the demand for automation increases users wants a more smarter and authentic solution which will not compromise on posts and time.

Despite the potential in Threads it lacks automations, drafts section where user can save their ideas for future use, detailed analysis that is required by the content creator on daily basis to examin and analyse their content and posts due to lack of automation it requires a lot of time to manage the profiles. Business and creators face diffuclty to maintain consistency, analyse performance and maximise their reach.

Thriend.io is a SAAS product that helps users manage their Threads social media platform. It schedules all posts to post regularly to maintain consistency. We will provide the draft section as Threads does not provide this feature alone so user can save their ideas there and can use them in future after refining them. Thriend.io will help users engage with other posts as our AI assistant will help to write comments through its extension. It will generate 1000 posts of Threads so the user doesn't have to think much for new ideas and posts. For a high number of posts sometimes it becomes a headache to schedule every single post so we recognize a feature of periodic posting in which a user just has to set the timer through which after every regular interval posts will be published. Thriend.io also provide advanced analytics to smartly plan the strategies for the future for better growth of Threads accounts

Looking ahead, Thriend.io aims to revolutionize Threads automation by incorporating predictive analytics, sentiment analysis, and advanced AI interactions to create an even more intelligent and adaptive experience. With continuous innovation, Thriend.io envisions becoming the go-to platform for social media automation, enabling users to stay ahead in the ever-evolving digital space.

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CHAPTER NUMBER 1

SOFTWARE PROJECT

MANAGEMENT PLAN

Chapter 1 Software Project Management Plan

1.1 Introduction:

This first chapter introduces the description of Thriend.io what is the problem statement, and how it will be designed and developed to solve that particular problem. This chapter will also explain the scope, requirements and objectives of the project.

1.2 Problem Statement:

Many users and influencers are doing their business and want to grow on different social media platforms. All they want is to grow their respective audience on specific social media platforms. Thread is among them and people are using Threads as it has a niche-specific audience so we can easily target our audience on Threads for their business growth they prefer Threads but the issue is we have to post on threads daily not just once in fact twice or thrice to keep up the good health of account not only this to grow here you also have to be very active and respond to other creator posts so that our account will grow and become more visible. But to remain active here takes a lot of time and sometimes we don't know how to respond to specific posts as we don't have those particular words. Not only this sometimes when new accounts start their journey they have to publish posts in massive amounts and scheduling each post is quite irritating so periodic posting will help to overcome this burden and schedule all the posts after regular intervals. For influencers, the analytics provided by the Threads itself is not enough as they want to analyze the audience daily or weekly so Thriend.io will also solve this problem by providing detailed analytics.

1.3 Project Description:

Thriend.io is a web-based application software that helps users generate multiple posts at a time, Do automated engagements with other users, Schedule multiple posts, and Publish posts periodically. Thriend.io also allows user to train their post generator that how the post will look and which tone it will use for reference Users can also provide it with some textual examples of others posts so that it will generate the posts according to it. It also provides the feature of drafts is in demand feature for threads users as they want to save their ideas somewhere and want to publish that after refinement in future. It helps users to see detailed analytics and make reports according to it so they can make more useful strategies.

1.4 Major Functionalities:

Major functionalities of this tool are:

1. Publish Threads posts directly from here
2. Schedule multiple posts
3. Schedule posts for a regular interval
4. Basic and advanced analytics
5. Post generation
6. AI Engagement assistance extension
7. Draft section – a place for ideas for future posting

1.5 Objective:

The main objectives of this SAAS are:

1. To overcome the time taken by users during handling Threads account
2. To maintain consistency
3. Think less work more
4. To grow Threads account and audience
5. To remain active on Threads
6. Providing a place to users where they can store their ideas

1.6 Tools and Techniques:

It is a web-based application and the following tools have been used in development:

1. React for frontend
2. Nodejs for backend
3. Express for Routings
4. MongoDB as a Database
5. Redis as a cache layer between a database
6. MS Word for Documentation
7. OPENAI API
8. Meta Threads API

1.7 Users:

This application will be used by the Threads audience only who will sign up/login to this web application they will generate their own post and publish them there will be no other role except Threads user.

1.8 Software Process Model:

Waterfall Model will be used for the development because:

1. It has very clear requirements

2. It is easy to manage due to the rigidity of the model.

1.9 Project Management Plan:

This section will elaborate on how the task will be divided among different people and resources. What will be the deliverables, risks and constraints?

1.10 Requirements Analysis Phase:

1.10.1 Requirements Identification:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
The initial step of this project is the identification of requirements which are functional and non-functional.	Requirements are collected and reviewed	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word **Hardware:** Laptop	None	None

Table 1.1 Requirements Identification Task

1.10.2 Define Use Cases:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
After clearing all requirements software architecture will start from Use case diagrams	Use Cases are written and then should be submitted as a Use-Case Diagram in Word format	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word, Draw.io **Hardware:** Laptop	Requirements must be clear.	Unclear requirements

Table 1.2 Defining Use Cases Task

1.10.3 Develop Domain Model:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
After we know about the use cases of a project we will start designing Domain Models for the system	Domain model will be reviewed	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word, Draw.io **Hardware:** Laptop	We must know the use cases before we move to the Domain Model	ill-defined use cases

Table 1.3 Domain Model Development Task

1.10.4 ERD Designing:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we make ERD Designs for the database	ERD Design will be reviewed	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word, Draw.io **Hardware:** Laptop	We must know the design of the domain model so we can see dataflow and what data we will need in this step	Any error in a Domain model

Table 1.4 ERD Designing Task

1.10.5 SRS Completion:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
This task includes making software requirements specification document	SRS document in Word or pdf form Requirement analysis completed	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word **Hardware:** Laptop	None	None

Table 1.5 SRS Completion Task

1.11 Designing Phase:

1.11.1 Develop Design:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
Development of architectural design	Architectural Diagram in PDF or Word format	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word **Hardware:** Laptop	Requirement analysis must be completed	Ill-defined requirements.

Table 1.6 Design Development Task

1.11.2 Develop Interfaces:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
This task includes how each screen looks like and what feature will come on what screen so we will make interfaces.	Figma Design, or Pdf format design	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word, Figma, Photoshop, Illustrator **Hardware:** Laptop	Architectural design must be completed	Ill-defined requirements, Use cases

Table 1.7 Interface Development Task

1.11.3 Class Diagram Designing:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we create a Class Diagram to show the flow of data and how they relate to each other.	Class Diagram in PDF or Word format	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word, Draw.io **Hardware:** Laptop	Architectural Design must be completed	ill-defined architectural design, and domain model.

Table 1.8 Class Diagram Task

1.11.4 Sequence Diagram Designing:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we make a Sequence Diagram to show the flow of data	Sequence Diagram in PDF or Word format	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word, Draw.io **Hardware:** Laptop	Architectural Design must be completed	ill-defined architectural design, and domain model.

Table 1.9 Sequence Diagram Task

1.11.5 Design Phase Verification:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we make sure that all the steps completed before the previous phase are correct. We do previous phase verification.	Design phase is completed it's document will be reviewed.	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word **Hardware:** Laptop	The whole Design phase must be completed	None

Table 1.10 Design Phase Verification Task

1.11.6 Software Test Documentation:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we define the test cases	Test cases are reviewed. Design phase is complete	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word **Hardware:** Laptop	Design phase should be complete	None

Table 1.11 Software Test Documentation Task

1.12 Implementation Phase:

1.12.1 System Implementation:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we develop the web application	Complete web application source files including images, videos or anything embedded in it submitted in zip.	**People**: Mohammad Shamil Khan, Dr Umar Rashid **Software:** MS Word, VS Code, Photoshop, Illustrator, MySQL, MongoDB **Hardware:** Laptop	The whole Design phase must be completed	None

Table 1.12 System Implementation

1.13 Gantt Chart

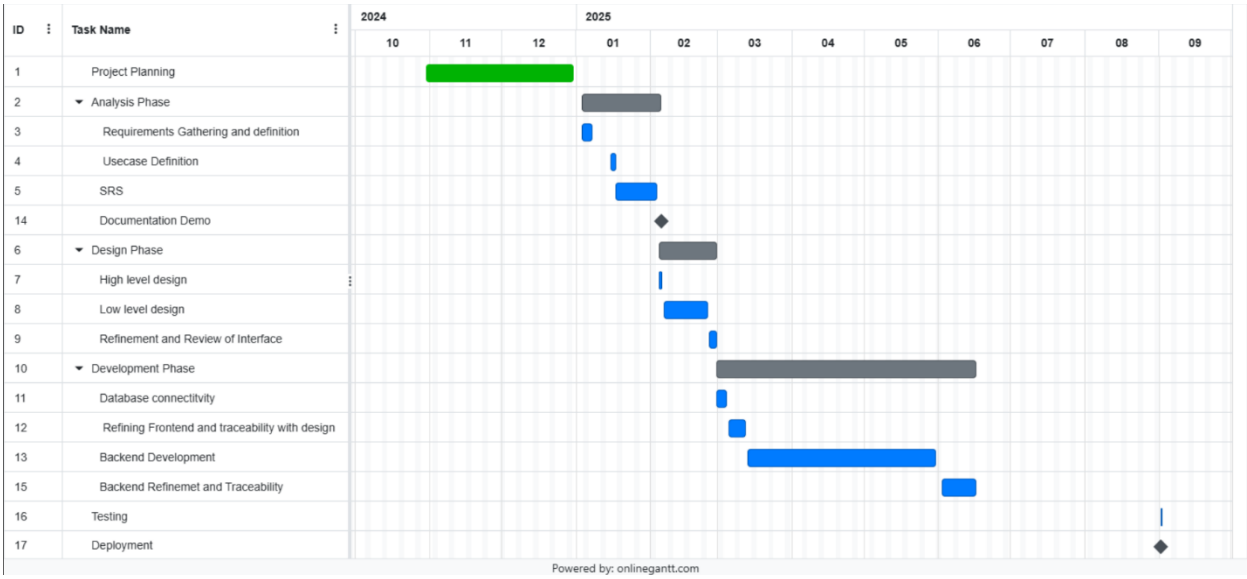

Figure 1 Gannt Chart

CHAPTER NUMBER 2

SOFTWARE REQUIREMENTS SPECIFICATIONS

2.0 Software Requirement Specification:

2.1 Introduction:

This chapter first describes the attributes of the software system. It highlights the user characteristics and major constraints of the tool. It also elaborates on the use cases and system sequence diagrams. Finally, this chapter explains the domain model and database design of the project.

2.1 Functional Requirements:

Functional requirements for this system are below:

- The system will provide a proper interface for all actors.
- Allow users to log in and sign up for software.

2.1.1 Post Publishing:

- Provide a user-friendly editor to write and format posts.
- Enable instant publishing directly on Threads.
- Provide features for uploading pictures, videos or media.
- Allow user to use AI assistance to write their singleton posts.
- Allow users to add child Threads or comment threads to the posts.
- Show users the preview of posts and how the post will look if specific text is written in a specific format.

2.1.2 Post Scheduling:

- Allow users to access dates and times to schedule a specific post.
- Select who can reply to user posts.

2.1.3 Regular Interval Scheduling (Future Release):

- Provide an option to the user to enter a niche and description
- Allow the user to select the time interval to publish the post.

2.1.3 Draft Management:

- Enable users to save unfinished ideas or posts
- Allow editing, organizing and tagging drafts for better organization.

2.1.4 Analytics:

- Provide metrics such as likes, engagement, comments, shares etc.
- Provide both collective and individual posts metrics.

2.1.5 Edit Post

- Allow users to edit the already existing post

2.1.6 Delete Post

- Allow users to delete the already existing post

2.1.7 Duplicate Schedule

- Allow user to duplicate and schedule the post

2.1.8 Edit Draft

- Allow users to edit the already existing post

2.1.9 Delete Draft

- Allow users to delete the already existing draft

2.2 Non-Functional Requirements:

2.2.1 Performance

- The system shall handle up to 10,000 scheduled posts per user per month without significant degradation in response time.
- Post scheduling and publishing actions should complete within 3 seconds.

2.2.2 Scalability

- The platform shall support scaling to accommodate 100,000 active users simultaneously without performance bottlenecks.

2.2.3 Availability

- The system shall have an uptime of 99.9%, ensuring minimal downtime for users.

2.2.4 Security

- The system shall encrypt all user data in transit using TLS 1.3 and at rest using AES-256 encryption.
- User authentication shall support multi-factor authentication (MFA).

2.2.5 Usability

- The platform's UI shall ensure users can perform core actions (e.g., scheduling a post) within 3 clicks or less.
- The system should support accessibility standards like WCAG 2.1 to cater to users with disabilities.

2.2.6 Maintainability

- The system shall allow updates and new feature deployments with no downtime for end-users through CI/CD pipeline

2.2.7 Responsiveness

- The platform shall provide a consistent user experience across devices (desktop, tablet, and mobile) and adapt to screen resolutions ranging from 720p to 4K.

2.2.8 Backup and Recovery

- All user data shall be backed up every 24 hours, and recovery processes shall ensure data is restored within 30 minutes in case of failure.

2.3 User Characteristics:

This tool can be used by all users of age greater than 18 as they can't make threads account until they are 18.

Users should have knowledge about:

- Basic web application
- Internet Literate

2.4 Constraints:

Users must have pc/laptop having Chrome downloaded. As this is a cloud application so it don't reserve much system resources.

2.5 Major Input and Output:

2.5.1 Major Inputs:

Some of the major Inputs of the system are:

1. Log in and Sign up credentials (username, email, password)

2. Posts Niche and description for AI assistant
3. Date and Time
4. Post text to publish
5. Audience type for AI training
6. Comment response
7. Comment response AI training

2.5.2 Major Outputs:

Some of the major outputs of a system are:

1. Analytics
2. Generated posts
3. Generated comments
4. Screen of all scheduled posts

2.6 Use Cases Diagram:

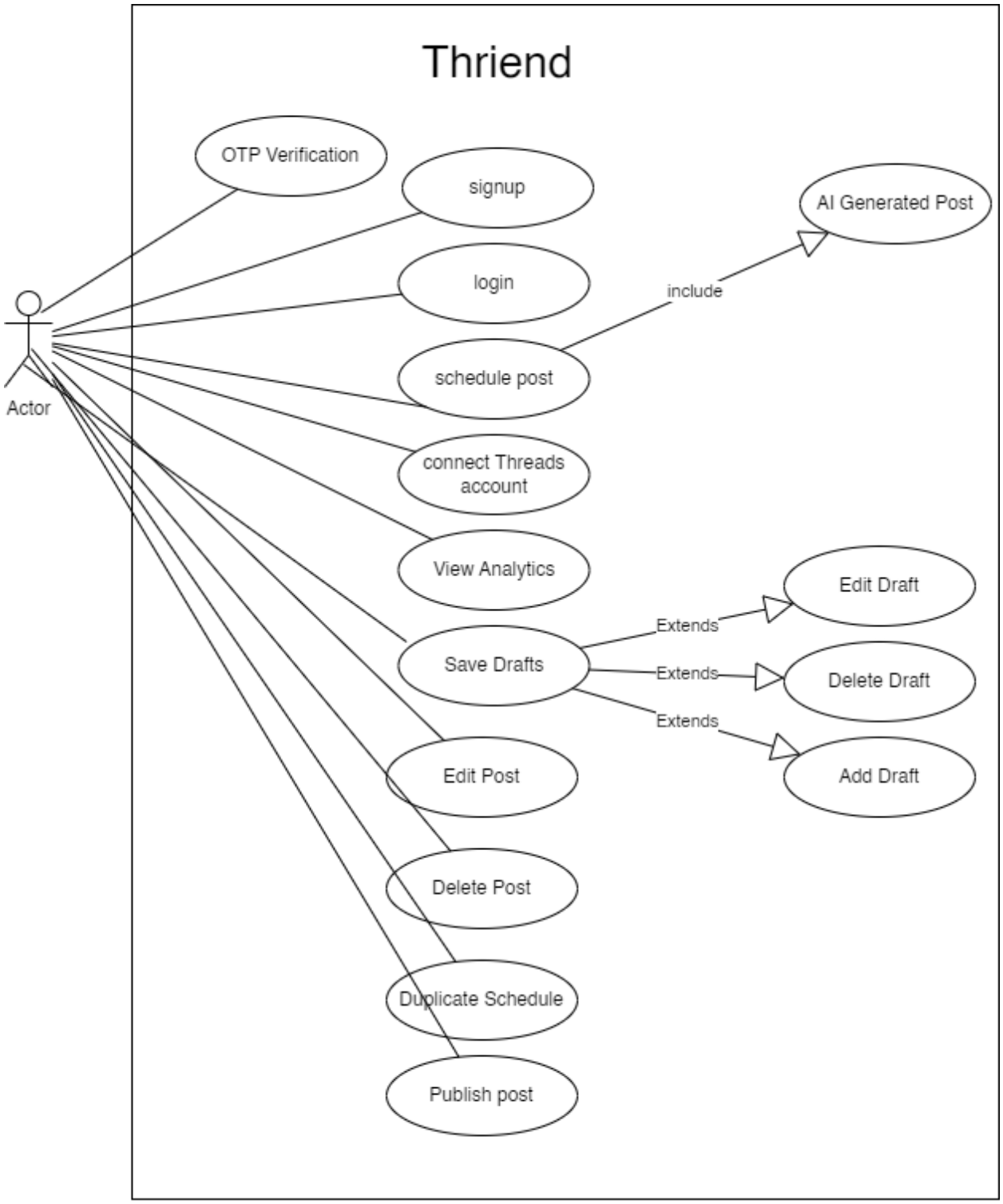

Figure 2.1 Use Case Diagram

2.7 Use Case Description:

2.7.1 Login

ID	UC1
Name	Login
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user has already signed up for an account The user knows his password
Post Condition	The user has logged in to the web application successfully. The system will redirect the user to connect Threads account page
Main Success Scenario	The user opens the application The system displays the login screen The user enters their credentials The system validates and shows the dashboard of the application
Alternative Flows or Extensions	**Server Down**: The user will wait until it gets reconnected **Enter Wrong Password:** System prompts to re-enter the password
Frequency	Almost frequently continuous

Table 2.1 Use case description of Login

2.7.2 Signup

ID	UC2
Name	Signup
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user must have connected to an internet
Post Condition	The user has signed up to the web application successfully. The system will redirect the user to connect Threads account page
Main Success Scenario	The user opens the application The system displays the signup screen The user enters their credentials
Alternative Flows or Extensions	**Server Down**: User will wait until it gets reconnected **Enter Wrong Password:** Enter incorrect password in confirm password field they will re-enter that password.
Frequency	Once per user

Table 2.2 Use case description of Signup

2.7.3 Connect Threads Account

ID	UC3
Name	Connect Threads Account
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user has already logged in for an account The user must have his Threads account
Post Condition	The user has connected their Threads account successfully The system has displayed the main dashboard page
Main Success Scenario	The user opens the application The system displays the connect account screen The user connects the Threads account by entering Meta Threads credentials The software validates and shows the dashboard of the application
Alternative Flows or Extensions	**Server Down**: The user will wait until it gets reconnected **Enter Wrong Password:** System prompts to re-enter the password
Frequency	After every 2 to 3 months as token expires

Table 2.3 Use case description of Connect Threads Account

2.7.4 Schedule Posts

ID	UC4
Name	Schedule Posts
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc The user has already logged in for an account The user must have his Threads account connected
Post Condition	The user has scheduled the post The system will publish the post as soon it approaches the scheduling time.
Main Success Scenario	The user opens the schedule post page. The system displays the calendar screen. The user will enter the post-data The user will set a time and date The user will publish the post for future
Alternative Flows or Extensions	**Server Down**: User will wait until it gets reconnected **Empty post:** System will disable the publish button
Frequency	Daily use

Table 2.4 Use case description of Schedule Post

2.7.5 View Analytics

ID	UC5
Name	View Analytics
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user has already logged in for an account The user must have his Threads account connected Must have followers greater than 100 to enable analytics
Post Condition	The user will be able to see all their analytics The user will be able to see daily, weekly or monthly analytics
Main Success Scenario	The user will select the timeframe Now the user will be able to see all analytics
Alternative Flows or Extensions	**Server Down**: The user will wait until it gets reconnected **Having followers less than 100:** The system will show a message
Frequency	Almost daily

Table 2.5 Use case description of View Analytics

2.7.6 Save Drafts

ID	UC6
Name	Save Drafts
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user has already logged in for an account The user must have his Threads account connected
Post Condition	Your Post and idea has been saved successfully in drafts
Main Success Scenario	Select the draft option Write anything and click save for future.
Alternative Flows or Extensions	**Server Down**: The user will wait until it gets reconnected
Frequency	Nearly daily

Table 2.6 Use case description of Save Drafts

2.7.7 Delete Drafts

ID	UC7
Name	Delete Drafts
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user has already logged in for an account The user must have his Threads account connected
Post Condition	Your Post and idea has been deleted successfully in drafts
Main Success Scenario	Select the draft option Write anything and click save for future.
Alternative Flows or Extensions	**Server Down**: The user will wait until it gets reconnected
Frequency	Nearly daily

Table 2.7 Use case description of Delete Drafts

2.7.8 Edit Post

ID	UC8
Name	Edit Post
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user has already logged in for an account The user must have his Threads account connected There must be a post exists in the list
Post Condition	Post Edited successfully
Main Success Scenario	Go to the post you want to edit Click the edit button Now edit or modify the post User will save the post Now the edited post have been save
Alternative Flows or Extensions	**Server Down**: The user will wait until it gets reconnected
Frequency	Multiple time per day

Table 2.8 Use case description of Edit Post

2.7.9 Delete Post

ID	UC9
Name	Delete Post
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user has already logged in for an account The user must have his Threads account connected Post must exists
Post Condition	Your post has been deleted successfully.
Main Success Scenario	Select the posts you want to delete Click delete and post will be deleted from the calendar
Alternative Flows or Extensions	**Server Down**: The user will wait until it gets reconnected
Frequency	Could be multiple time per day.

Table 2.9 Use case description of Delete Post

2.7.10 Publish Now

ID	UC10
Name	Publish Now
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user has already logged in for an account The user must have his Threads account connected Post must exists
Post Condition	Your post has been published successfully.
Main Success Scenario	Write a post add images or videos if needed Publish post now
Alternative Flows or Extensions	**Server Down**: The user will wait until it gets reconnected
Frequency	Could be multiple time per day.

Table 2.10 Use case description of Publish Post

2.7.11 Duplicate Schedule

ID	UC11
Name	Schedule duplicate
Primary Actor	Threads user
Pre-Condition	The user has a laptop, pc or mobile The user has already logged in for an account The user must have his Threads account connected Post must exists
Post Condition	Your post has been duplicate successfully. Your post has been scheduled
Main Success Scenario	Write a post add images or videos if needed Publish post now
Alternative Flows or Extensions	**Server Down**: The user will wait until it gets reconnected
Frequency	Could be multiple time per day.

Table 2.11 Use case description of Duplicate schedule

2.8 System Sequence Diagrams

An interaction diagram which shows the interactions between the external actor and the system and the events generated by these actors is called system sequence diagrams

2.8.1 Login

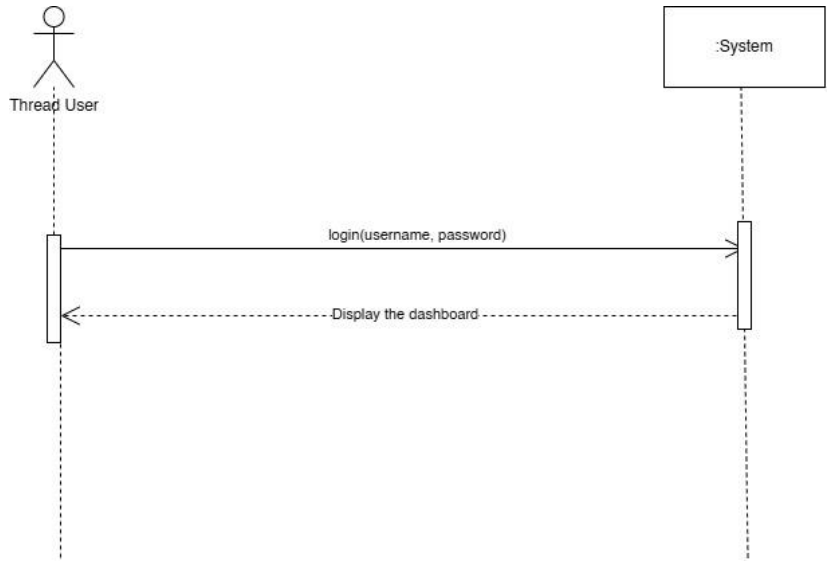

Figure 2.1 Login SSD

2.8.2 Signup

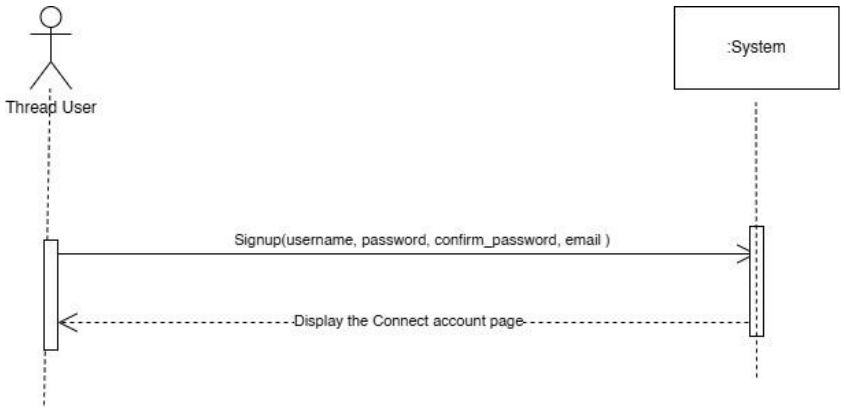

Figure 2.2 Signup SSD

2.8.3 Analytics

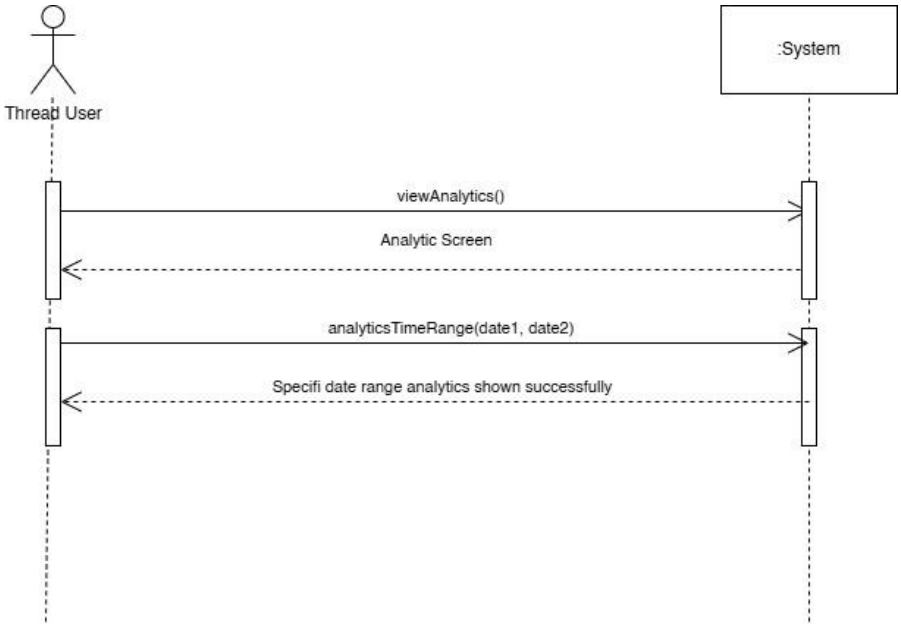

Figure 2.3 Analytics SSD

2.8.4 Delete Post

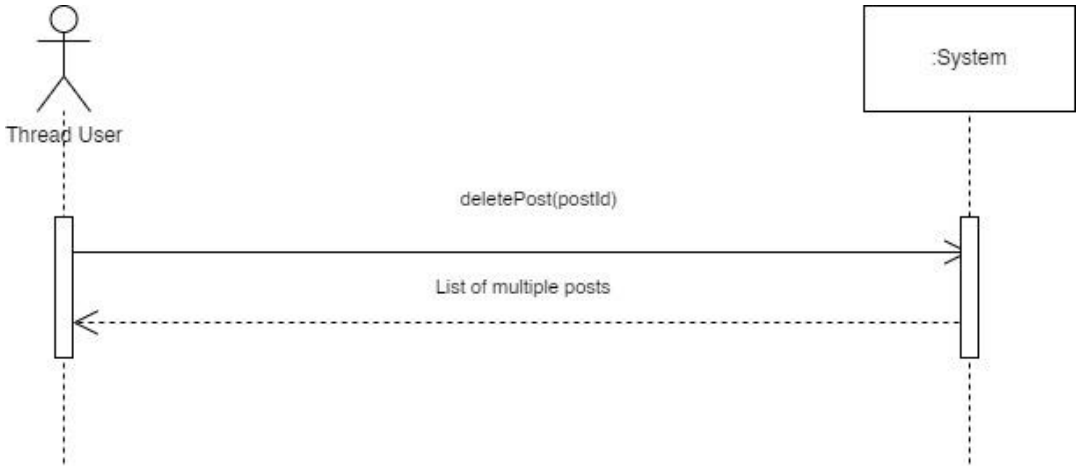

Figure 2.4 Delete Post SSD

2.8.5 Edit Post

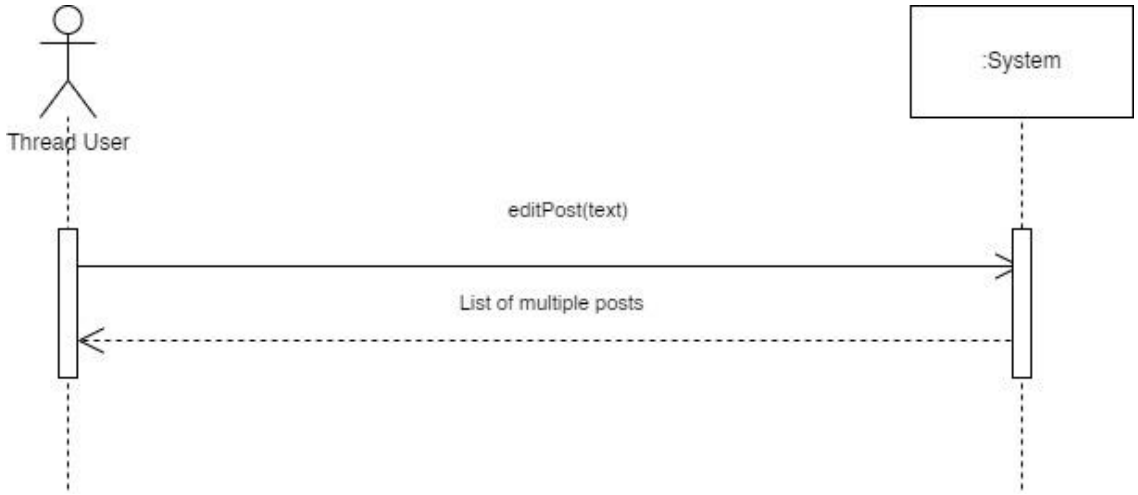

Figure 2.5 Edit Post SSD

2.8.6 Save Draft

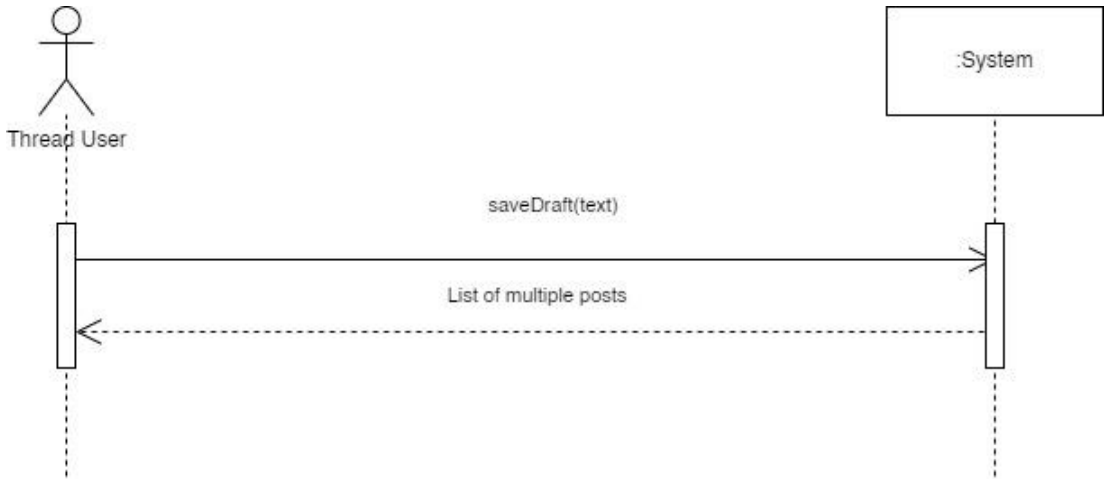

Figure 2.6 Save Draft SSD

2.8.7 Delete Draft

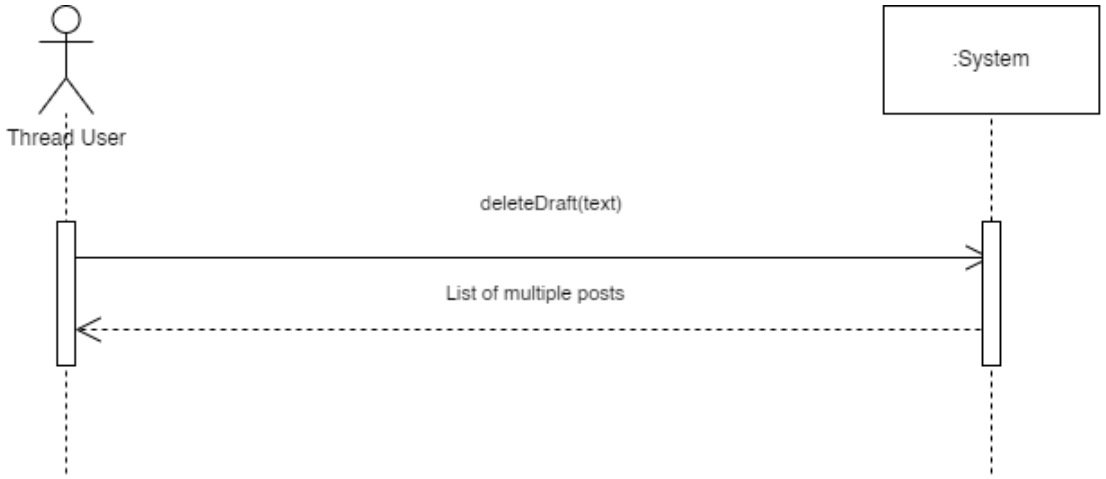

Figure 2.7 delete Draft SSD

2.8.8 Publish Now

Figure 2.8 Publish Now SSD

2.8.9 Schedule Post

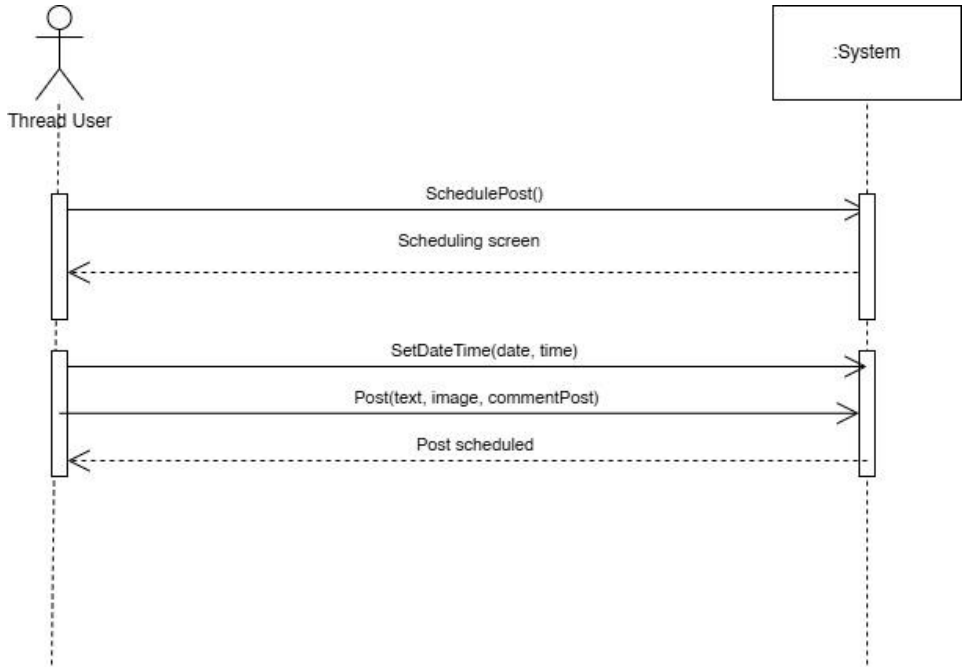

Figure 2.9 Schedule Post SSD

2.8.10 Duplicate Schedule

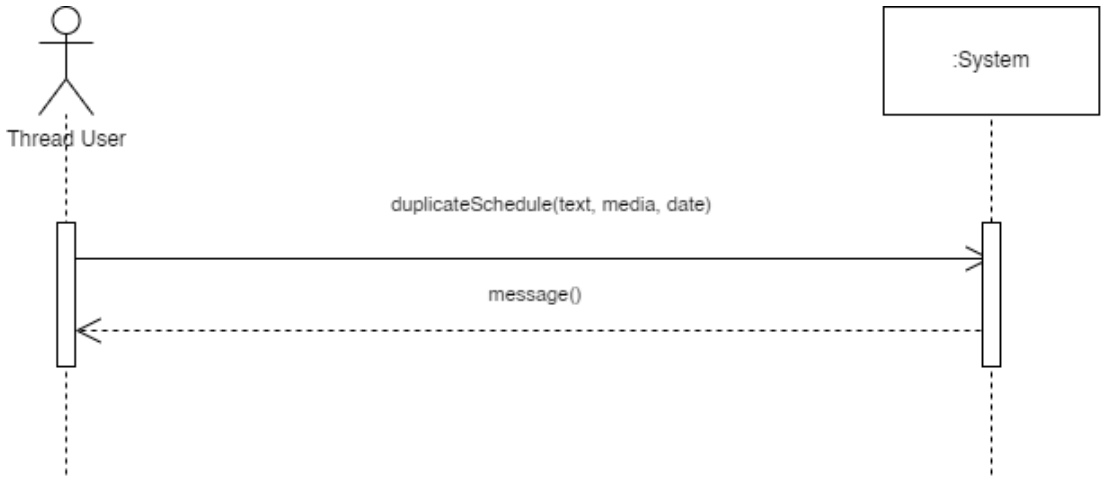

Figure 2.10 Duplicate Schedule SSD

2.9 Domain Model

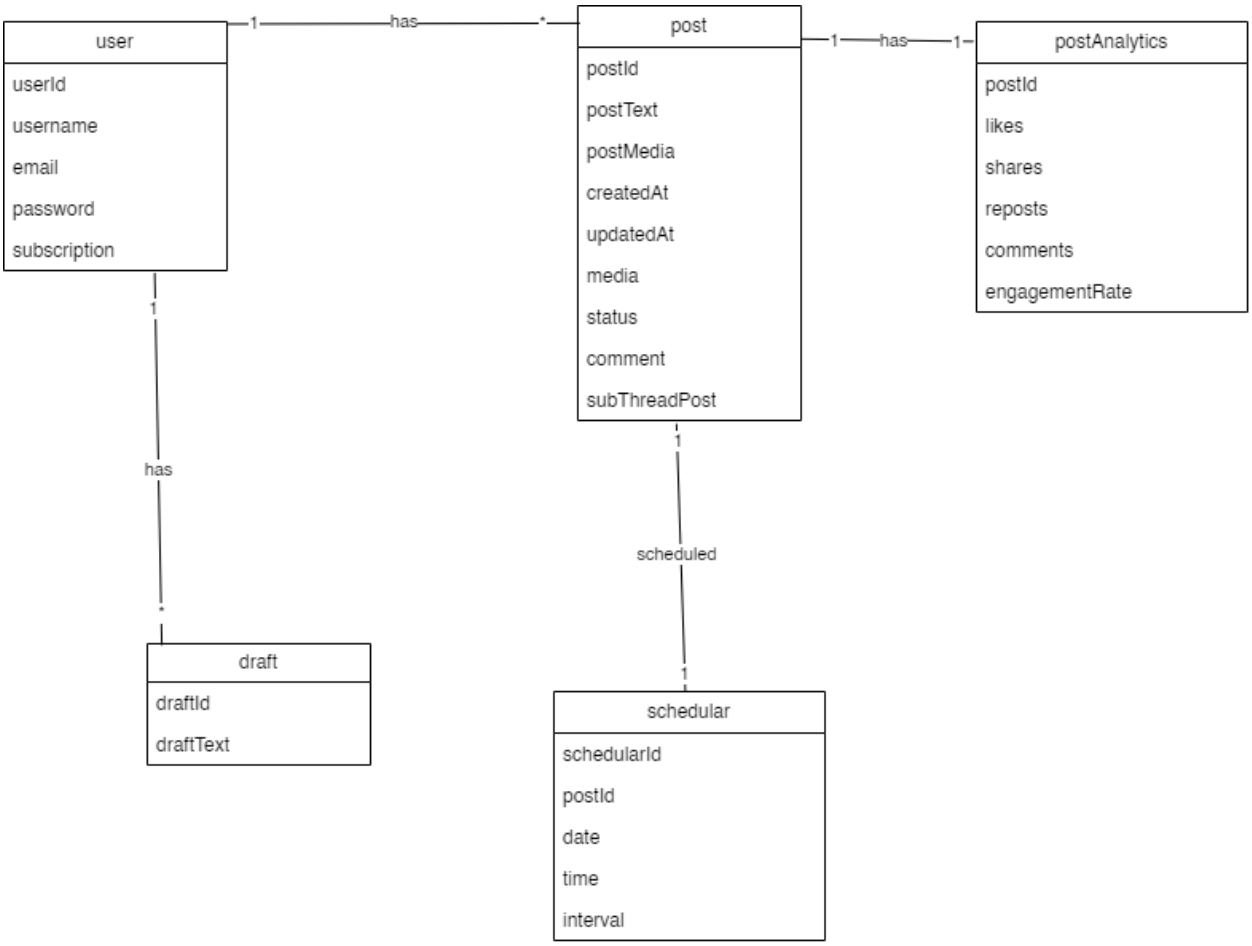

Figure 2.11 Domain Model

2.10 Entity Relationship Model

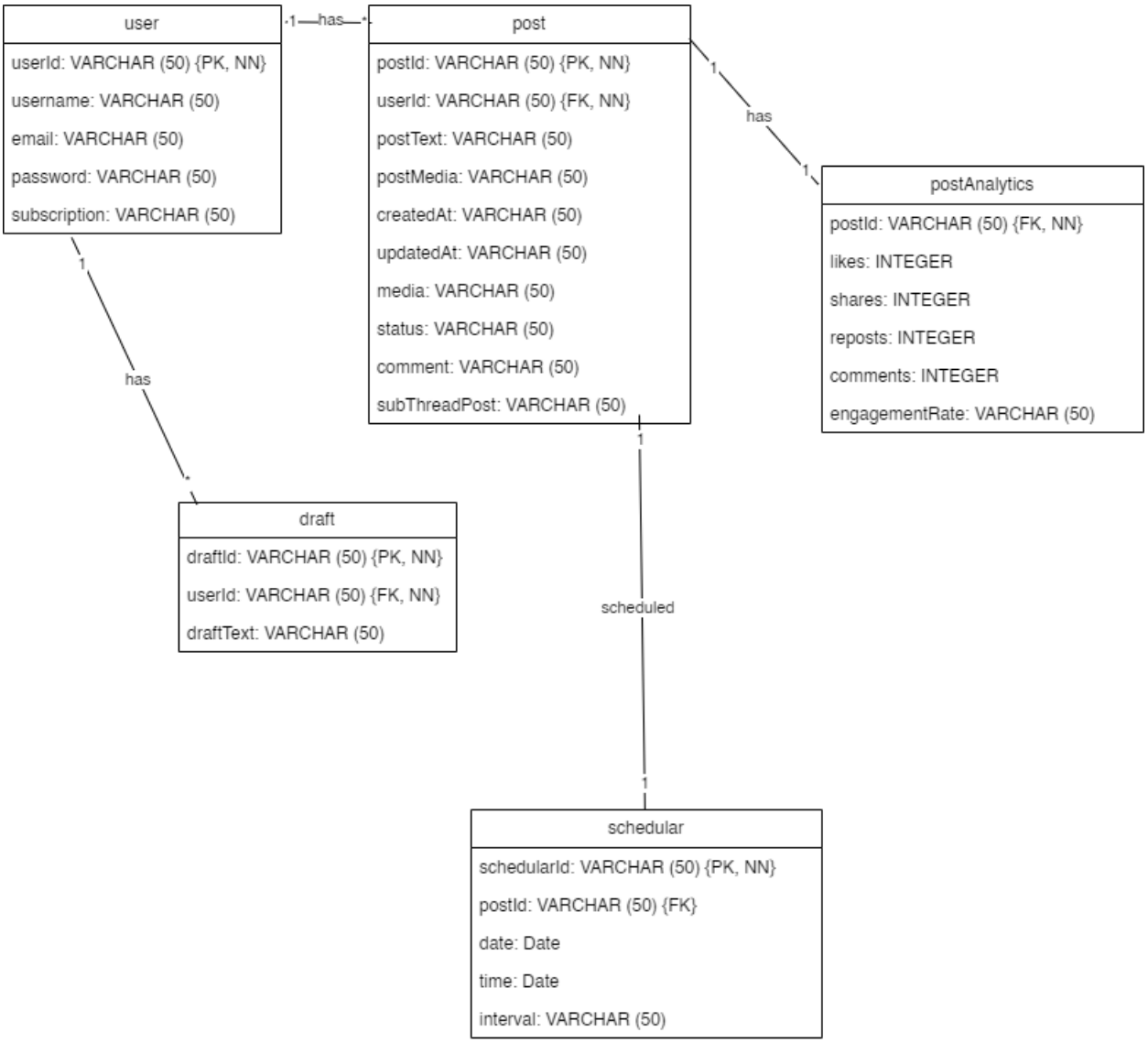

Figure 2.12 ERD – Entity Relationship Model

CHAPTER NUMBER 3

SOFTWARE DESIGN

DESCRIPTION

3.1 Introduction:

Software Design Description SDD is a representation of a software design used to communicate system design information to all stakeholders. It shows how the software system will be structured to satisfy the requirements.

3.2 Sequence Diagrams

A sequence diagram is an interaction diagram that shows how objects operate with one another and what order. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and class involved in the scenario and the sequence of messages exchange between the objects needed to carry out the functionality of the scenario. Sequence diagrams are developed against the use cases.

3.2.1 Login

This diagram is showing that when a user will enter login credentials how system will interact with each other and finds a user from the database.

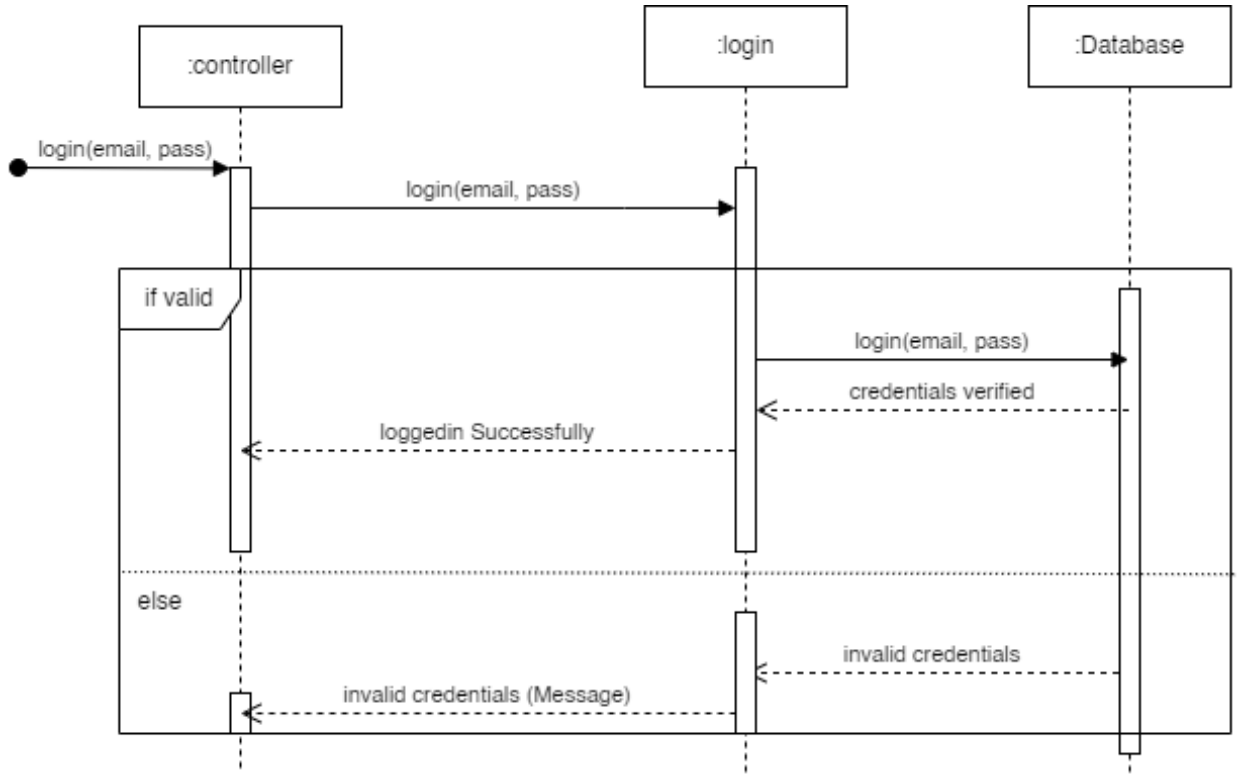

Figure 3.1 Login Sequence Diagram

3.2.2 Signup

It is showing what happens if user already exist or how will a user will signup

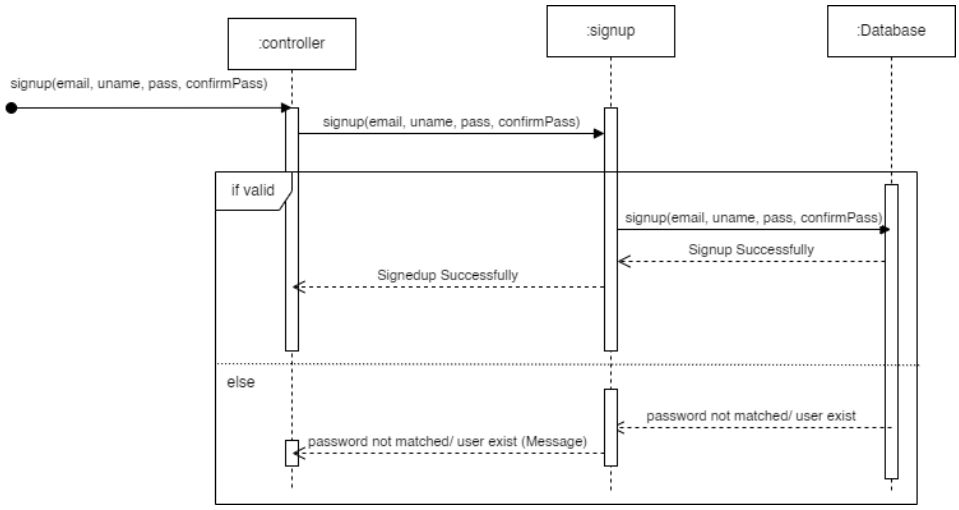

Figure 3.2 Signup Sequence Diagram

3.2.3 Schedule Post

In schedule post user will schedule the post by setting up the date, time, enter the text and image to be scheduled.

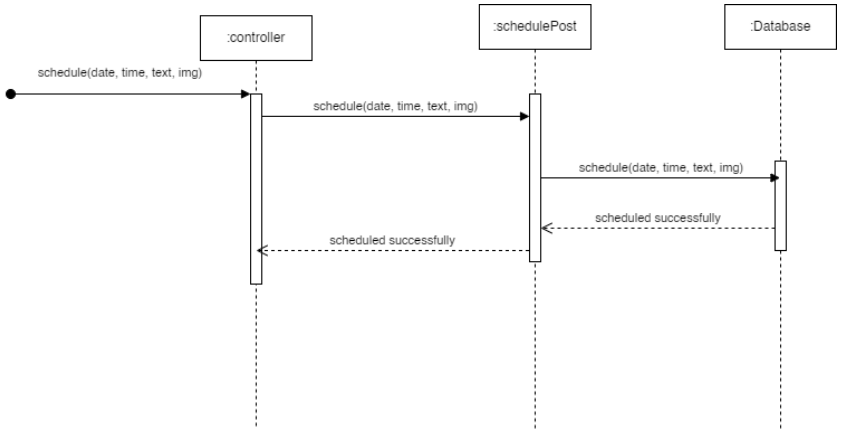

Figure 3.3 Schedule Post Sequence Diagram

3.2.4 Connect Threads Account

In connect threads account if a user has already logged in on chrome with Threads account then during the connection user will directly redirected to dashboard without asking for Threads credentials but if he is not he must enter the credentials of Meta Threads account i.e the username and password.

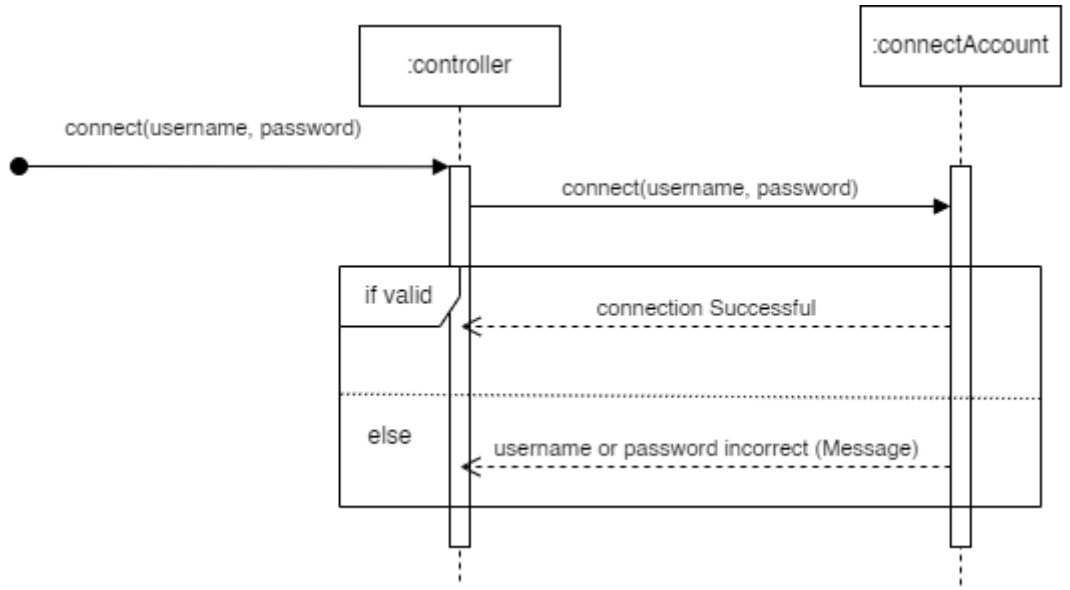

Figure 3.4 Connect Threads Account Sequence Diagram

3.2.5 View Analytics

In view analytics user can view all the analytics of their posts and account.

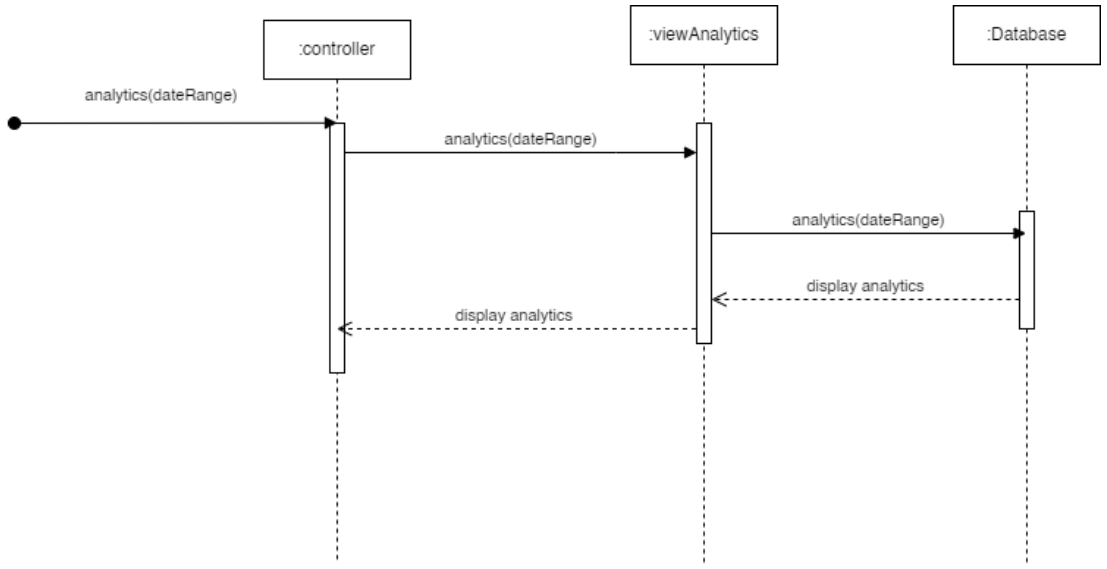

Figure 3.5 View Analytics Sequence Diagram

3.2.6 Drafts

User can save their ideas and drafts in a database and can easily view all the saved data and drafts in a draft section and the draft section will also have edit, delete and add feature.

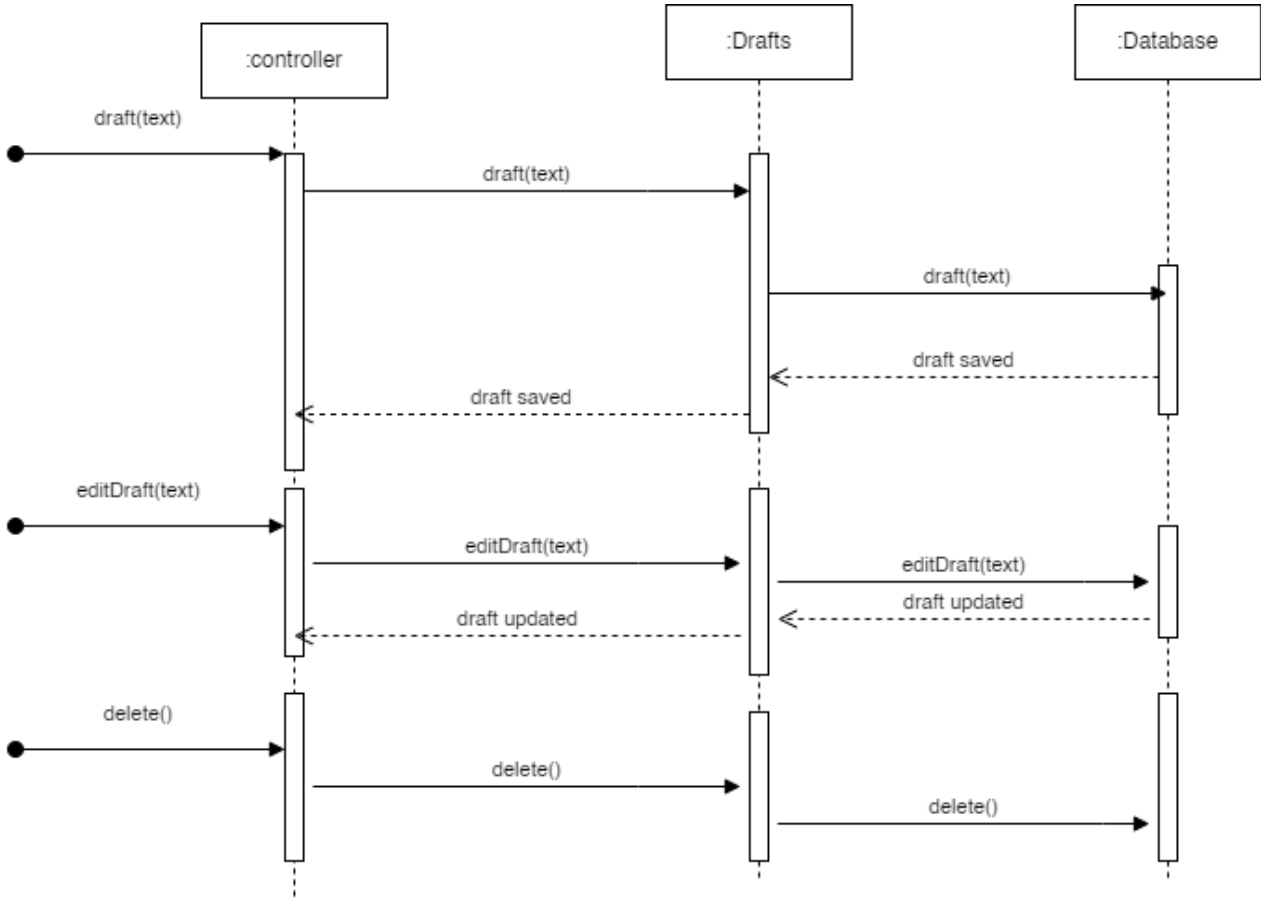

Figure 3.6 Drafts Sequence Diagram

3.2.7 Publish Now

In publish now user will directly write and add media and can publish the post from thriend software.

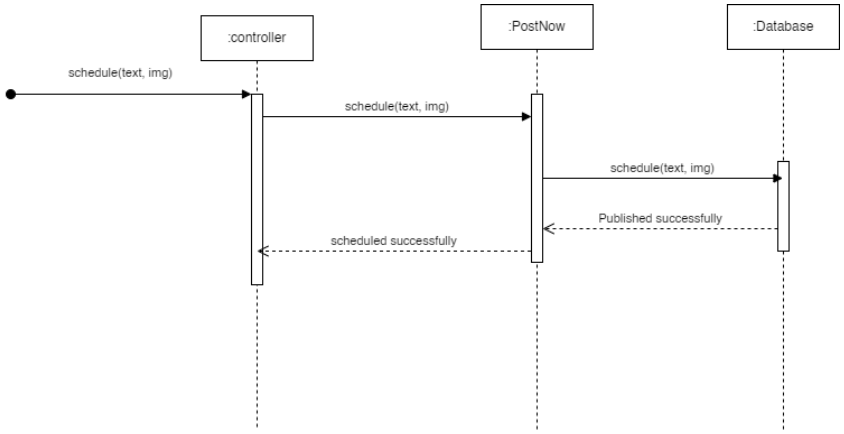

Figure 3.7 Schedule Post Sequence Diagram

3.3 Class Diagram

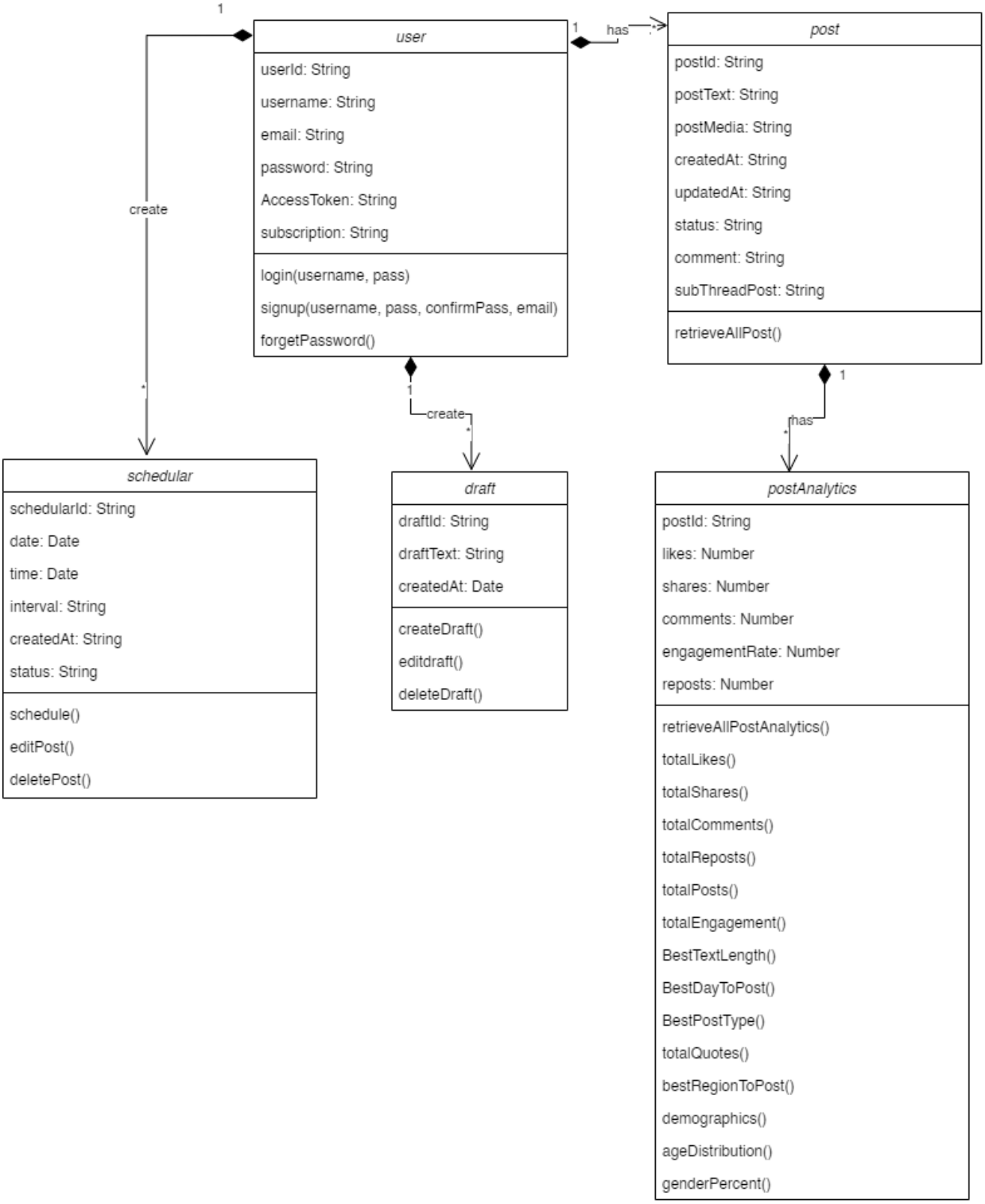

Figure 3.8 Class Diagram

3.4 Interface Design:

It is the process in which we create the prototype of the screen images. In this section, we draw the interface based on our requirements. And later during the implementation, we will use them for actual development.

3.4.1 Login Interface

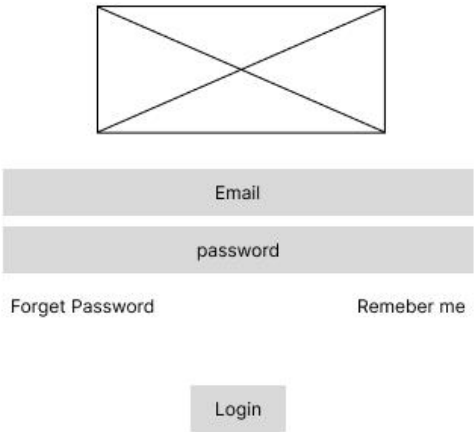

Figure 3.13 Login/Signup Interface

3.4.2 Connect Account

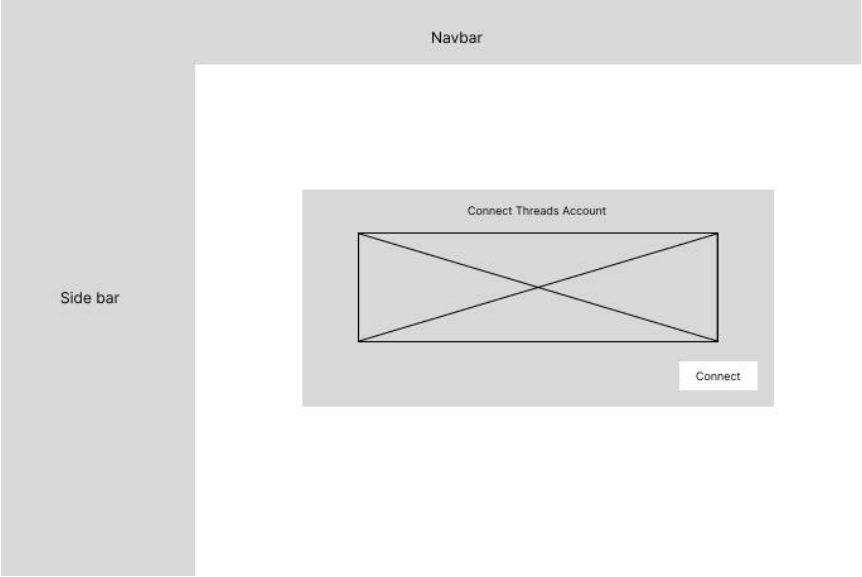

Figure 3.14 Connect Account

3.4.3 Scheduler

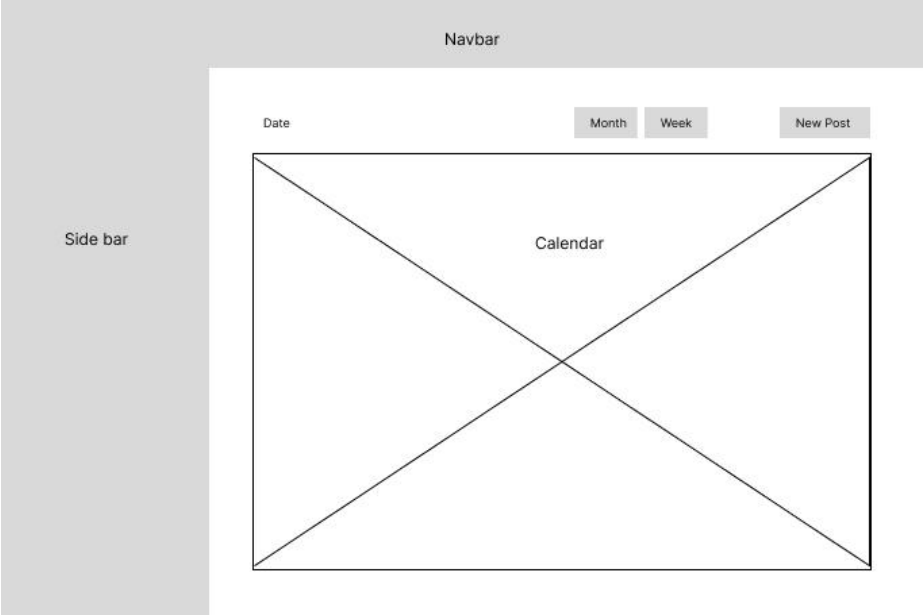

Figure 3.15 Schedular

3.4.4 Create Post

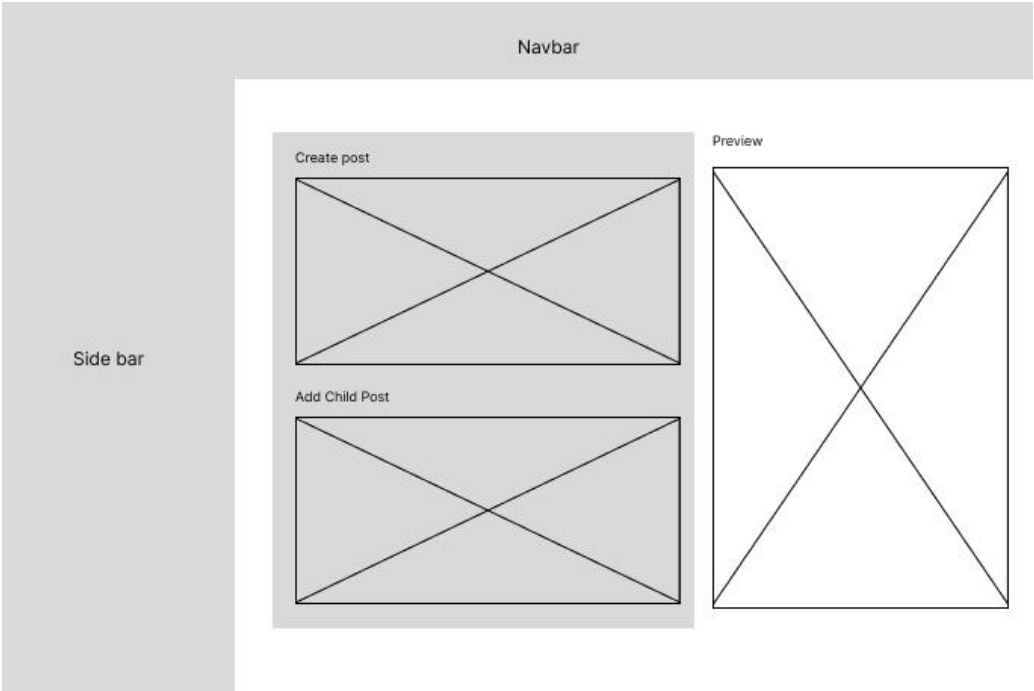

Figure 3.16 Create Post View

3.4.5 Delete Post

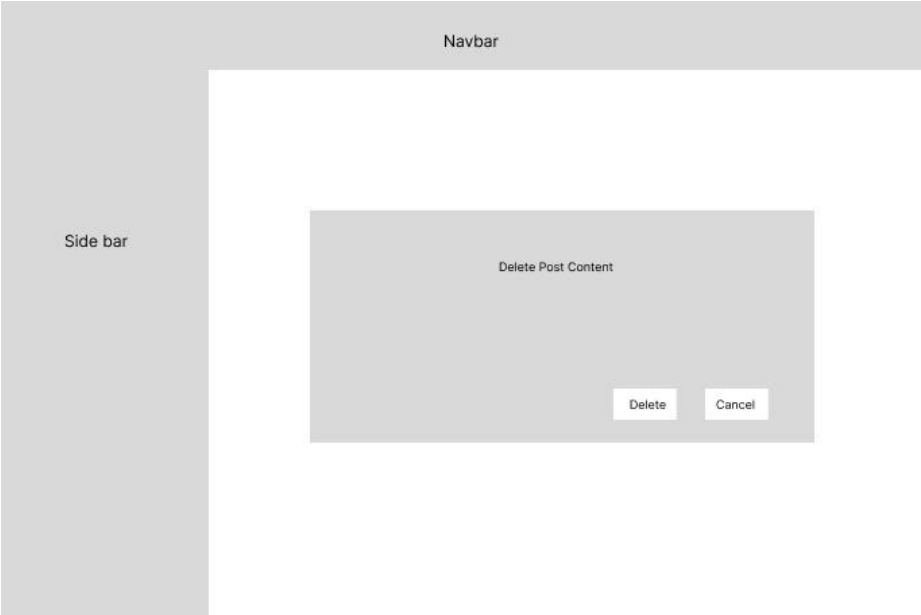

Figure 3.17 Delete Post View

3.4.6 Recurring Post

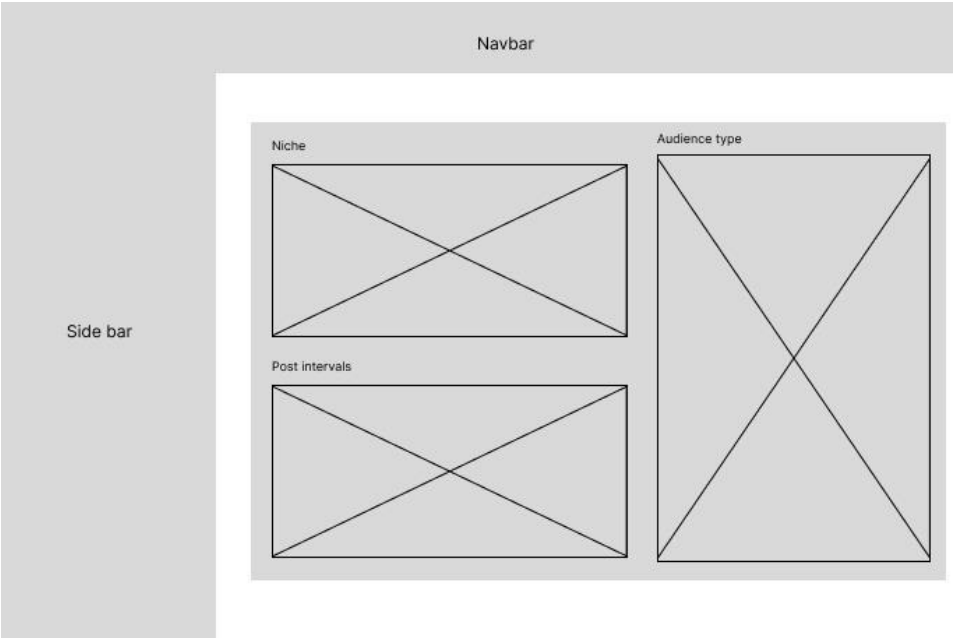

Figure 3.18 Schedule Recurring Post View

3.4.7 Post Generator

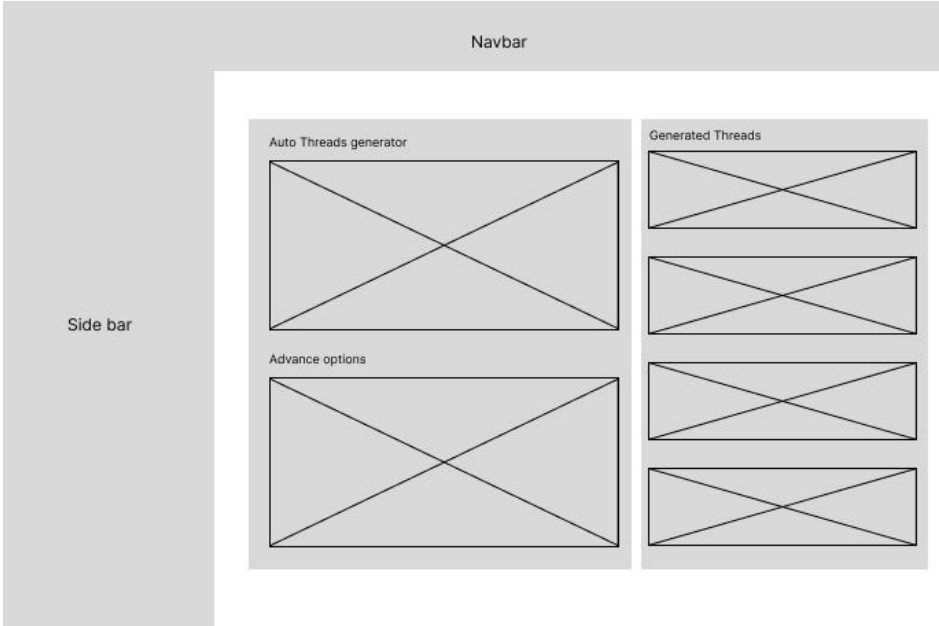

Figure 3.19 Post Generator View

3.4.8 ATM Credentials

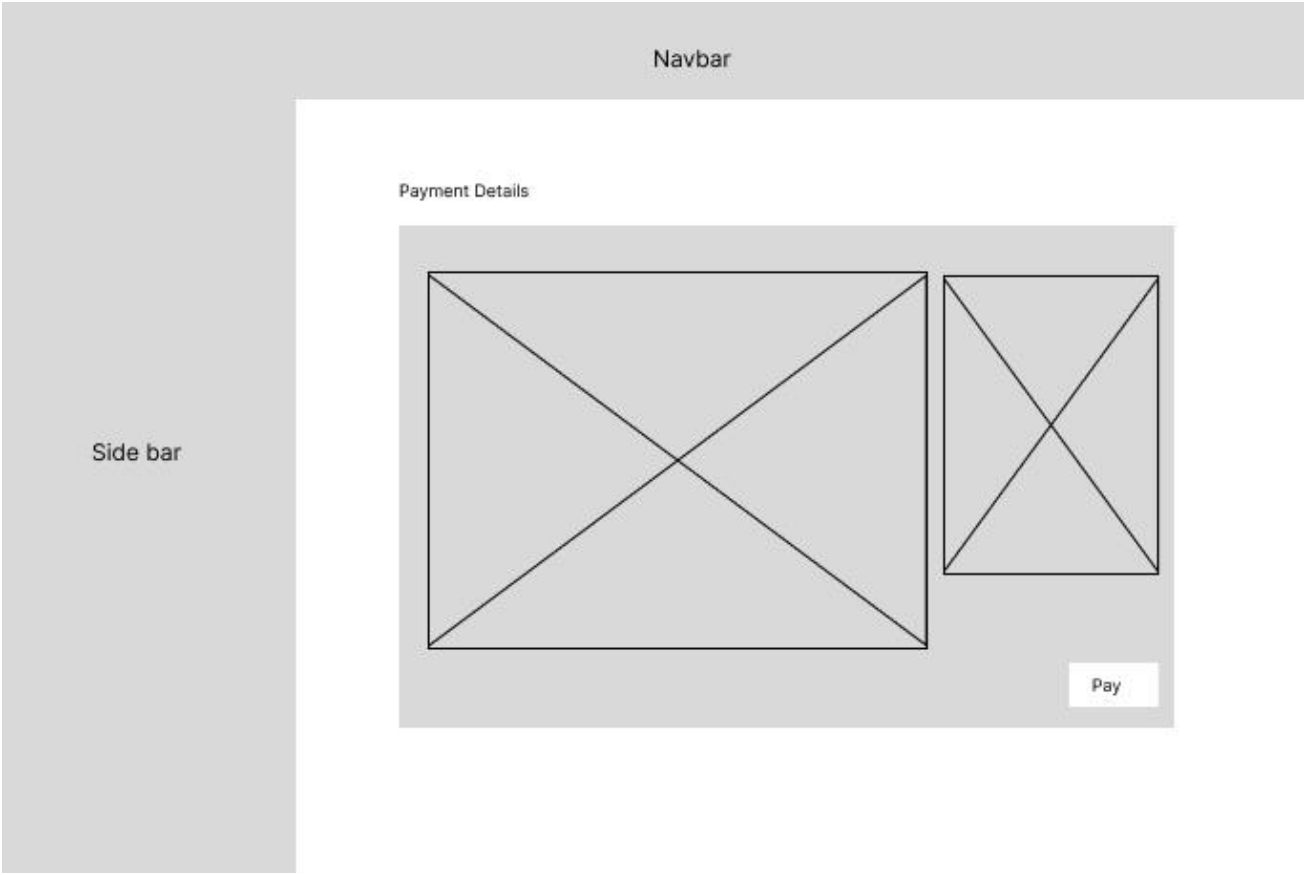

Figure 3.20 ATM Credentials View

3.4.9 Reply Management

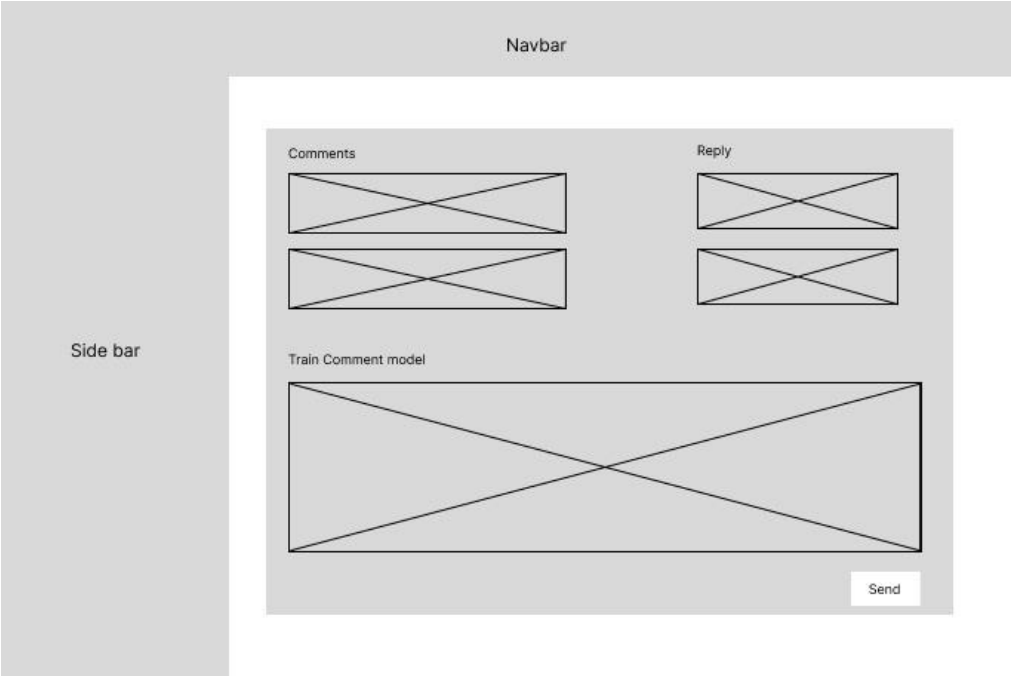

Figure 3.21 Reply Management View

3.4.10 Analytics

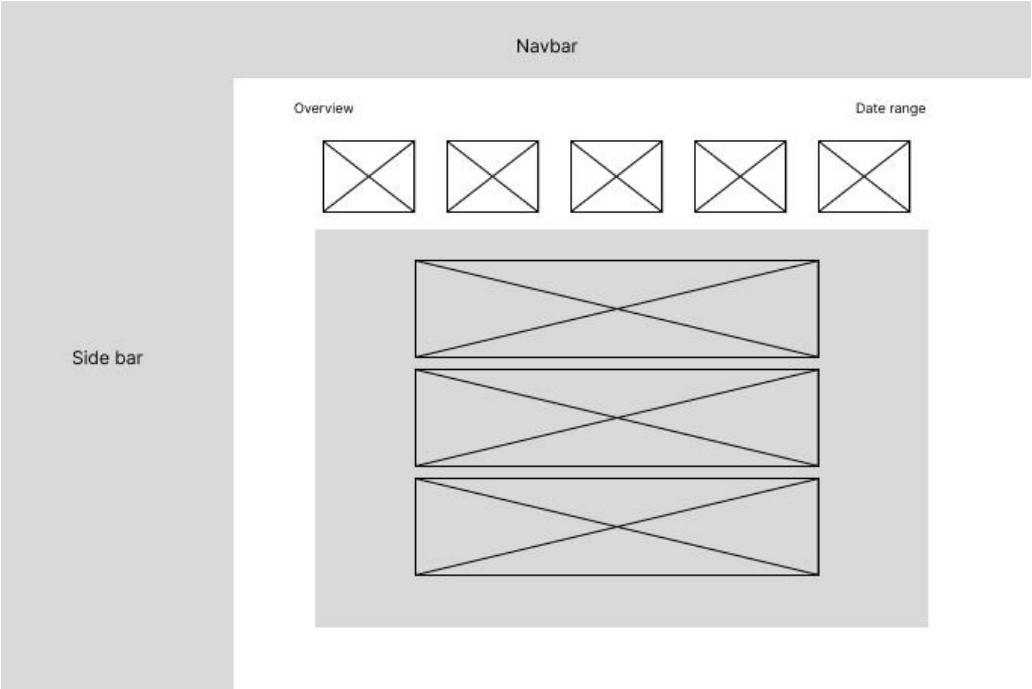

Figure 3.22 Analytics View

3.4.11 Login/signup

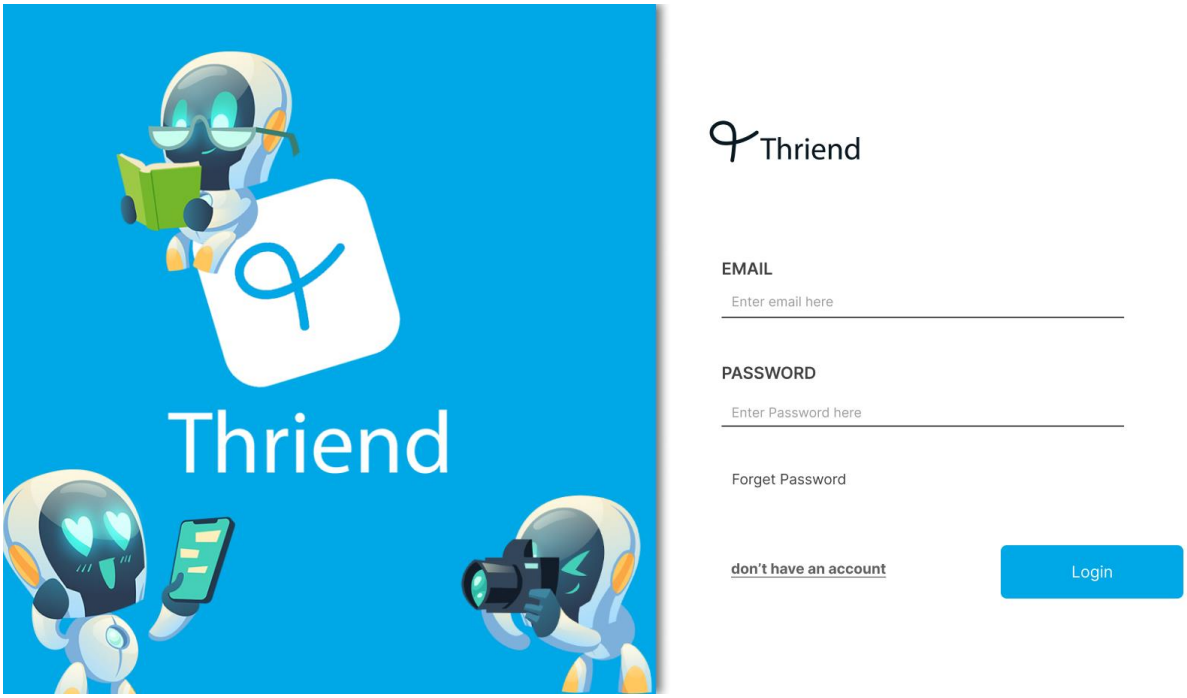

Figure 3.4.11 Login/Signup

3.4.12 Menu

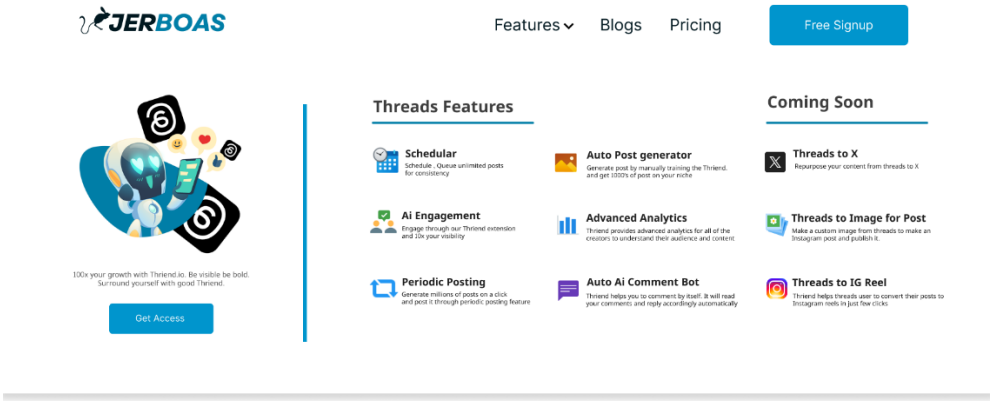

Figure 3.4.12 Menu

3.4.13 Landing page

Figure 3.4.13 Landing Page

3.4.14 Pricing

Figure 3.4.14 Pricing

3.4.15 Email Sent

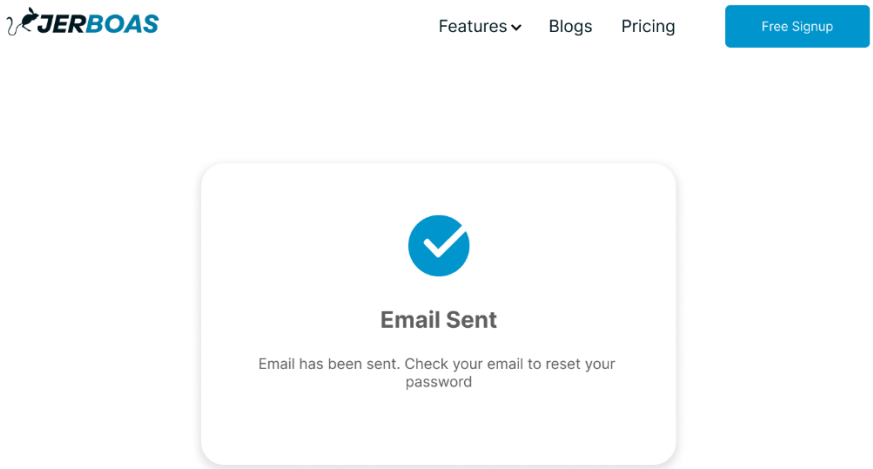

Figure 3.4.15 Email Sent

3.4.16 Calendar

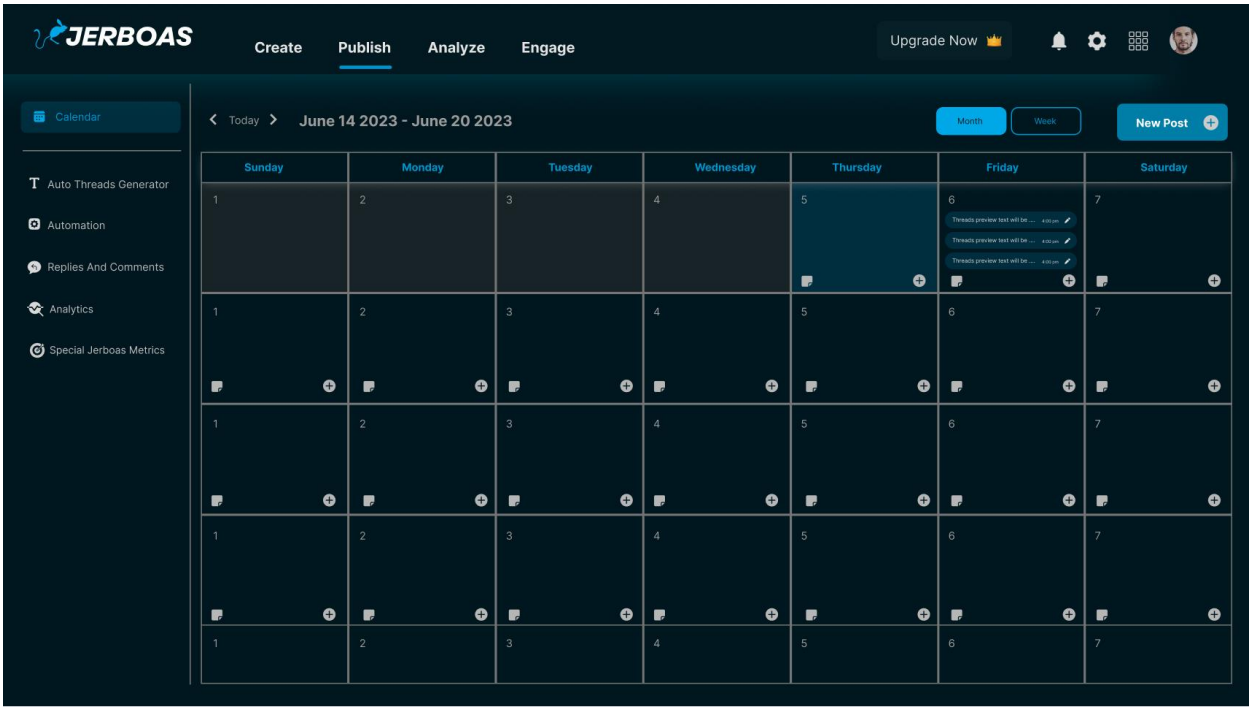

Figure 3.4.16 Calendar

3.4.17 Calendar Weekly view

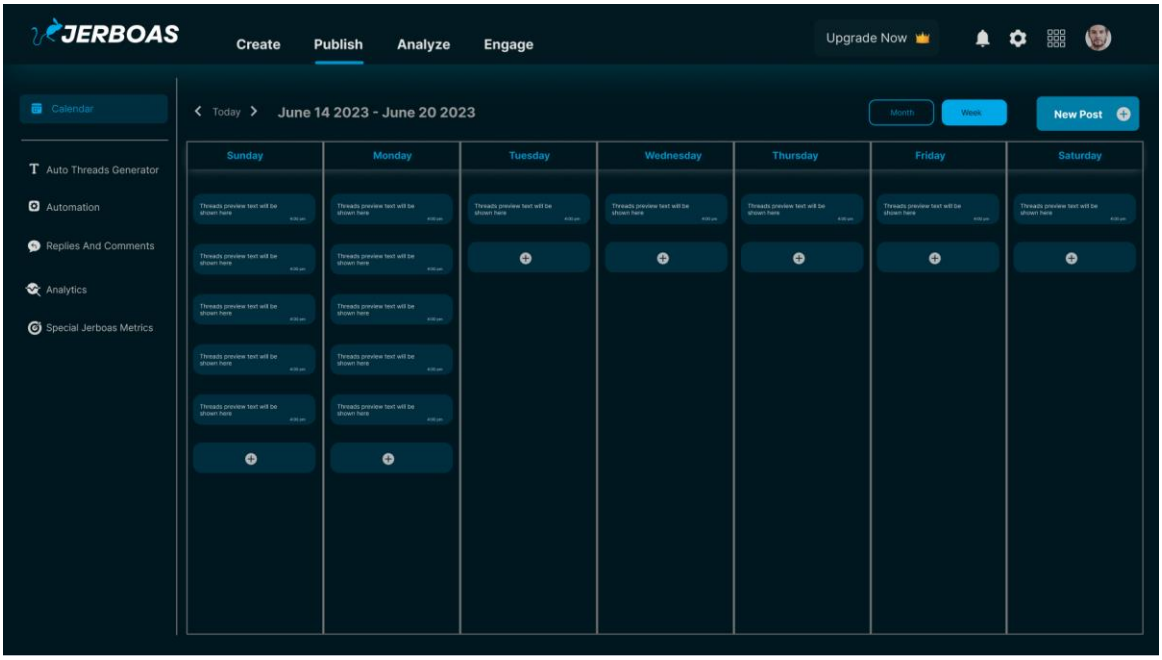

Figure 3.4.17 Calendar Weekly View

3.4.18 Connect Threads Account

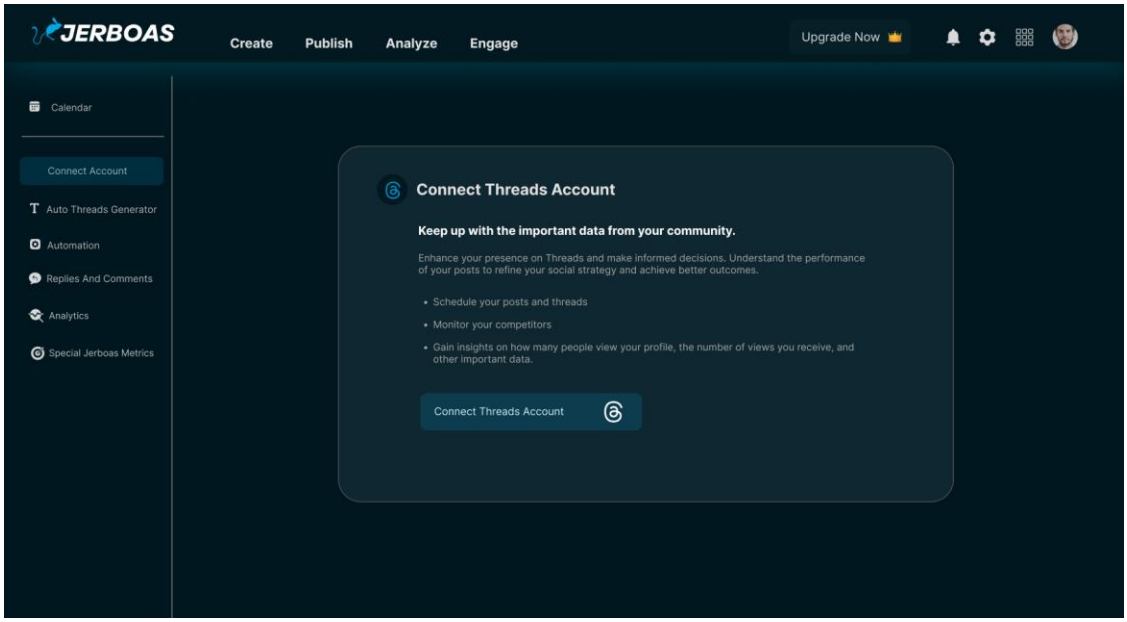

Figure 3.4.18 Connect Threads Account

3.4.19 Analytics

Figure 3.4.19 Analytics View

3.4.20 Create Post

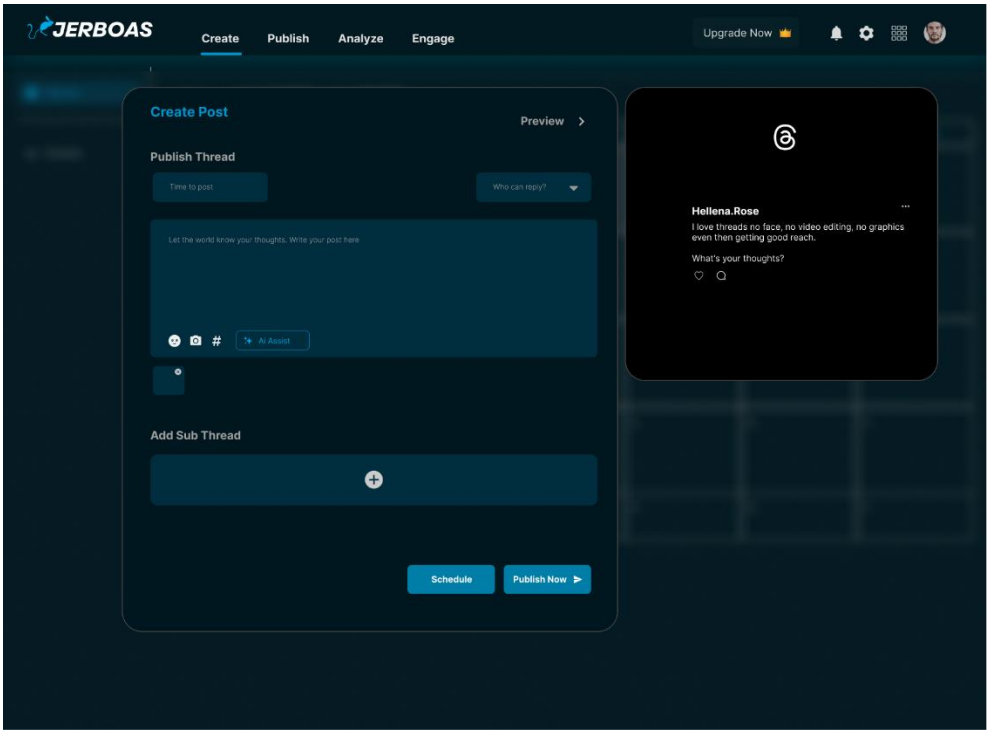

Figure 3.4.20 Create Post

3.4.21 Delete Post

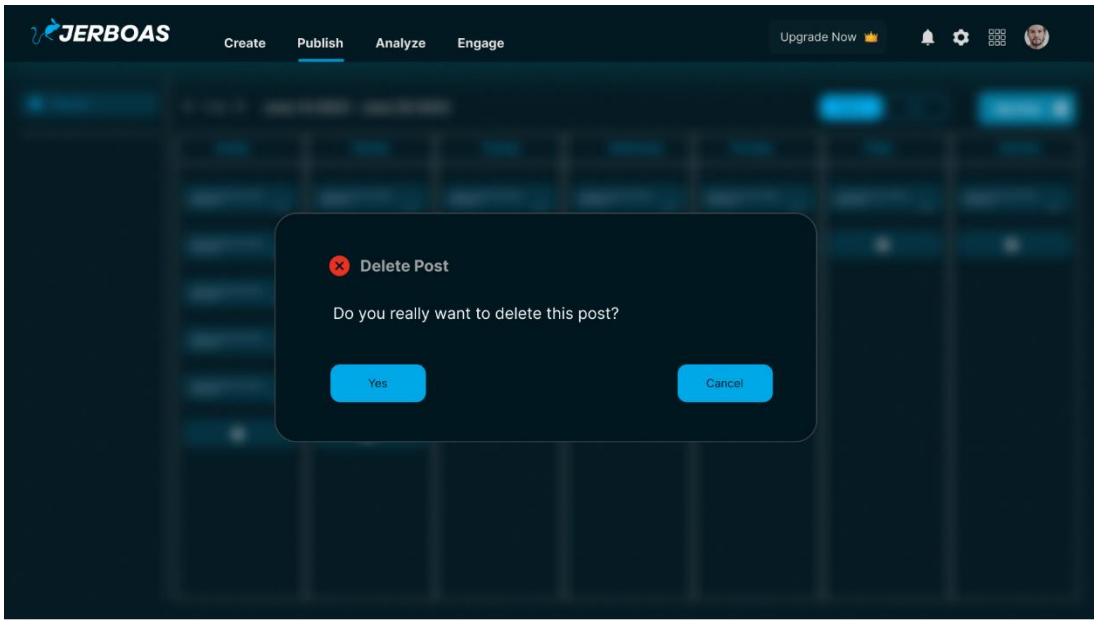

Figure 3.4.21 Delete Post

3.4.22 Reply Management

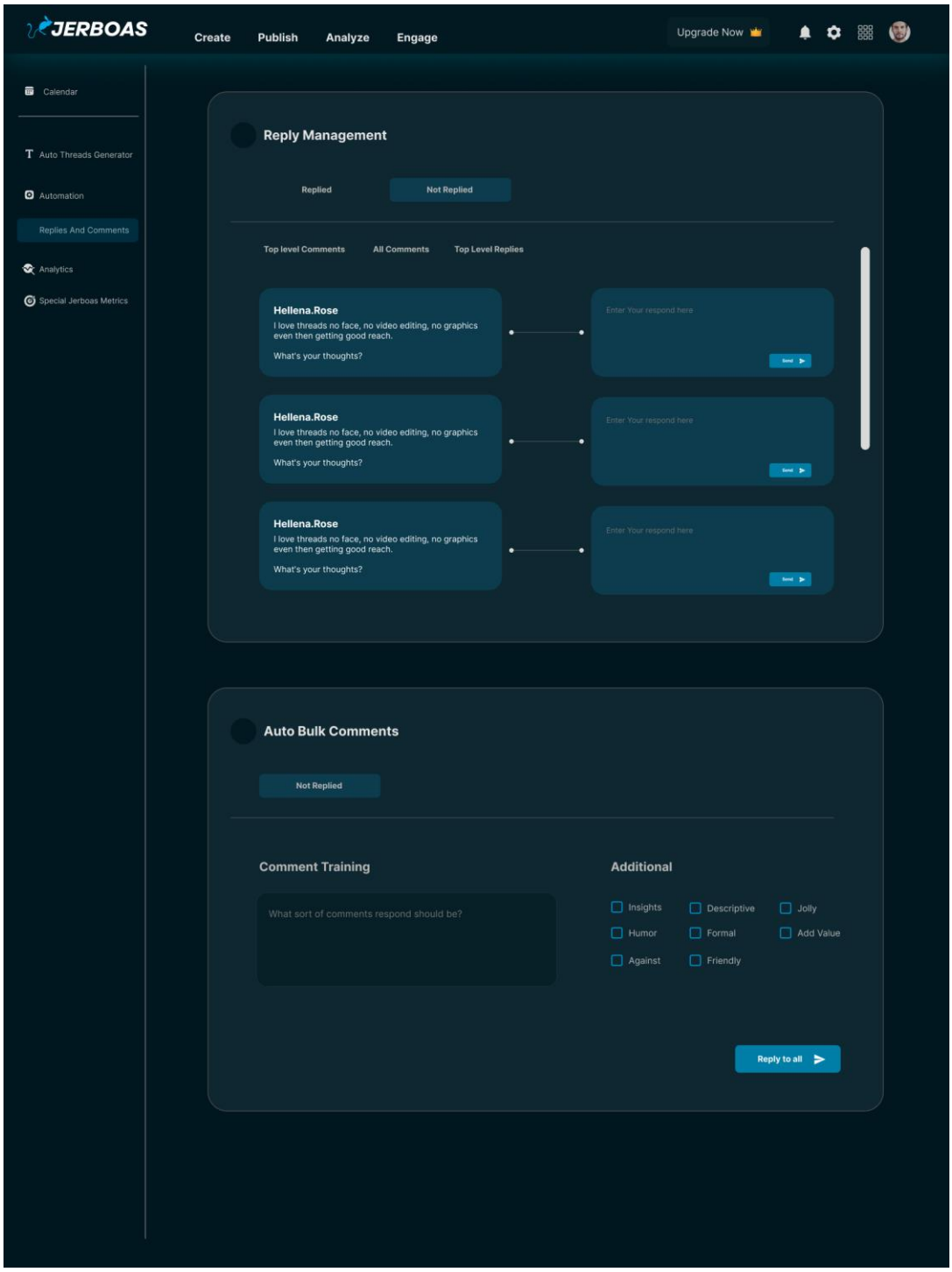

Figure 3.4.22 Reply Management

CHAPTER NUMBER 4

SOFTWARE TESTING

4.1 Introduction

Software test document is a type of document under which test will determine whether system under test satisfies requirements or work correctly. Developing test cases can also help find problems in the requirement or design of an application.

4.2 Test plan

Test planning is an activity that ensures that there is a list of tasks and milestones in a baseline plan to track the progress of the project. Test plan determines the scope and the risks that needed to be tested and are not to be tested. Deciding fail and pass criteria.

4.3 Features For Testing

Features to be tested are following:

- Login
- Schedule Post
- Analytics
- Generate AI Comments
- Generate Content
- Drafts
- Buy Subscription

4.4 Test Cases

4.4.1 Login

ID	T1
Description	The user is registered and login into account
Tester	User
Setup	User starts the website
Instructions	• Select Login option • Enter email as mskeditor@gmail.com and password as “admin” • Select Login
Expected Results	Dashboard displayed
Actual Result	Dashboard is displayed
Status	Test Passed

Table 4.1 Test Case for Login

4.4.2 Schedule Post

ID	T2
Description	Verify that a user can successfully schedule a post on Threads for a future date and time
Tester	User
Setup	• User must be logged into the Threads scheduling tool • Ensure the user has a valid Threads account connected.
Instructions	• Navigate to the Schedule Post section. • Enter the post content (text, images, or videos). • Select a future date and time for the post. • Click the Schedule button. • Verify if the post appears in the Scheduled Posts list. • Wait until the scheduled time passes. • Check if the post is successfully published on Threads.
Expected Results	• The post is successfully saved in the Scheduled Posts list. • At the scheduled time, the post is automatically published on Threads. • The user receives a confirmation or notification upon successful posting.
Actual Result	Post is scheduled
Status	Test Passed

Table 4.2 Test Case for Schedule Post

4.4.3 Analytics

ID	T3
Description	Verify that the analytics feature correctly tracks and displays engagement metrics (likes, reposts, replies, quotes, and views) for a posted Thread.
Tester	User
Setup	• User must be logged into the Threads analytics tool (e.g., Thriends.io). • Ensure the user's Threads account is connected and has at least one posted Thread. • The analytics feature must be enabled and accessible.
Instructions	• Navigate to the Analytics Dashboard. • Select a previously posted Thread from the list. • Check if the following metrics are displayed: • Likes count • Replies count • Reposts count • Quotes count • Views count • Interact with the post manually (e.g., like, reply, repost) or wait for real interactions. • Refresh the analytics page and check if the metrics update in real-time or after a short delay. • Verify if percentage trends (increase/decrease) are displayed correctly compared to previous data.
Expected Results	• The analytics dashboard correctly displays all engagement metrics. • Metrics update accurately based on real-time or scheduled sync. • Percentage trends (up/down) are displayed correctly.
Actual Result	Analytics has been displayed
Status	Test Passed

Table 4.3 Test Case for Analytics

4.4.4 Add Drafts

ID	T4
Description	Verify that users can successfully save, edit, and publish draft posts in the Threads scheduling tool
Tester	User
Setup	• User must be logged into the Threads scheduling tool. • Ensure the user's Threads account is connected. • The draft feature must be enabled.
Instructions	• Navigate to the Create Post section. • Enter text, images, or a video for the post. • Click the Save as Draft button. • Navigate to the Drafts section and confirm the saved post appears. • Click on the draft and edit the content. • Click Save again and check if changes are updated.
Expected Results	• The post is successfully saved in the Drafts section. • Users can edit and update drafts without issues.
Actual Result	Drafts created
Status	Test Passed

Table 4.4 Test Case for Drafts

4.4.5 Edit Draft

ID	T5
Description	Verify that users can successfully edit an existing draft post in the Threads scheduling tool
Tester	User
Setup	• User must be logged into the Threads scheduling tool. • Ensure the user's Threads account is connected. • At least one draft post should already exist.
Instructions	• Navigate to the Drafts section. • Select an existing draft post. • Update the post • Click Save to update the draft. • Reopen the same draft to confirm that changes have been applied
Expected Results	• The draft post is updated with the new content. • The changes persist when the draft is reopened.
Actual Result	Drafts Edited
Status	Test Passed

Table 4.5 Test Case for Edit Draft

4.4.6 Delete Draft

ID	T6
Description	Verify that users can successfully delete a draft post in the Threads scheduling tool
Tester	User
Setup	• User must be logged into the Threads scheduling tool. • Ensure the user's Threads account is connected. • At least one draft post should already exist.
Instructions	• Navigate to the Drafts section. • Select an existing draft post. • Click the Delete option for the draft. • Confirm the deletion when prompted. • Refresh or revisit the Drafts section to verify removal.
Expected Results	• The selected draft post is permanently removed from the Drafts section. • Deleted drafts do not reappear after refresh.
Actual Result	Drafts Deleted
Status	Test Passed

Table 4.6 Test Case for Delete Draft

4.4.7 Edit Post

ID	T7
Description	Verify that a user can successfully edit a scheduled post before its scheduled publishing time
Tester	User
Setup	• User must be logged into the Threads scheduling tool • Ensure the user has a valid Threads account connected • At least one scheduled post must already exist
Instructions	• Navigate to the Scheduled Posts section. • Select an existing scheduled post. • Modify the post content (text, images, or videos) or update the scheduled date and time. • Click the Save/Update button. • Reopen the scheduled post and verify that the changes are reflected. • Wait until the updated scheduled time passes. • Check if the updated post is published correctly on Threads.
Expected Results	• The scheduled post is updated with the new content and/or rescheduled time. • Changes are visible when reopening the scheduled post. • At the updated scheduled time, the post is published correctly.
Actual Result	Post Edited
Status	Test Passed

Table 4.7 Test Case for Edit Post

4.4.8 Delete Post

ID	T8
Description	Verify that a user can successfully delete a scheduled post before it is published
Tester	User
Setup	• User must be logged into the Threads scheduling tool • Ensure the user has a valid Threads account connected • At least one scheduled post must already exist
Instructions	• Navigate to the Scheduled Posts section. • Select an existing scheduled post. • Click the Delete option for the post. • Confirm the deletion when prompted. • Refresh or revisit the Scheduled Posts section to verify removal. • Wait until the original scheduled time passes. • Check if the deleted post does not appear on Threads.
Expected Results	• The scheduled post is permanently removed from the Scheduled Posts list. • Deleted posts are not published on Threads at the scheduled time.
Actual Result	Post Deleted
Status	Test Passed

Table 4.8 Test Case for Delete Post

4.4.9 Publish Now

ID	T9
Description	Verify that a user can successfully publish a post immediately on Threads using the Publish Now
Tester	User
Setup	• User must be logged into the Threads scheduling tool • Ensure the user has a valid Threads account connected
Instructions	• Navigate to the Create Post section. • Enter the post content (text, images, or videos). • Click the Publish Now button. • Verify if the post appears immediately in the Published Posts list within the tool. • Check on Threads to confirm that the post is live instantly.
Expected Results	• The post is published immediately without delay. • The post appears in the Published Posts list. • The content is visible on Threads right away.
Actual Result	Post Published
Status	Test Passed

Table 4.9 Test Case for Publish Now Post

CHAPTER NUMBER 5

CONCLUSIONS AND FUTURE ENHANCEMENTS

Introduction

This document describes the project conclusions and future enhancements i.e what type of new features can be added with time.

Summary and Conclusions

This project helps all the users active on social media mainly who want to sell something like courses or any digital stuff online but don't have much time to remain active and engage with an audience 24/7 and It also takes a lot of time so this project is helping users to save their time and maintaining consistency through different features like scheduling, engagements, auto content generation and save ideas in drafts.

This is a web based application but it can be extended for android users in the next few years.

Future Enhancements

In future the application can be enhanced by:

- IOS Applications
- Android Application
- Integrate twitter for more ease to switch their whole account to Threads by transferring all the post and schedule it on Threads and also by posting a same post on both platform simultaneously.
- Addition of Advance Analytics feature
- Addition of AI Generated Posts, comments features
- Addition of Recurring post scheduling feature
- Addition of Payment gateways feature
- Integrate Instagram for more convenient sharing of stuff between Threads and Instagram.
- By providing a feature through every user can make their threads bot with no code here.

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